Supplementary Material: Detailed Nonlocal Regularization, Extended Cross-Validation, Code Verification, Reproducibility Bindings and Public-Data Assessment

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S1. Nonlocal damage regularization of the full-field reference

The main text (Section "Nonlocal damage regularization of the full-field reference") reports only the headline conclusion of this study; the full route selection, implementation verification, tables and diagnostics are collected here. The full-field-reference section of the main paper quantifies the surrogate's bias against full-field references, but it leaves two things open: the in-house full-field reference is itself a local-damage model, so its own mesh dependence is unmeasured, and the strain localization at which the coupled FFT iteration diverges is a boundary of the reference rather than of the surrogate. This study addresses both by adding a strain-localization regularization to the in-house three-dimensional CPFEM reference and measuring what it does and what it does not do. All runs use the locked material and protocol of the reference (Ti-6Al-4V at 0° unless stated, $S_0 = 0.32\,\mathrm{MPa}$, $D_c = 0.42$, damage threshold $p_D = 0.05$, $n_{\mathrm{sub}} = 30$, strain rate $10^{-3}\,\mathrm{s}^{-1}$, seed 42, $\Delta\varepsilon = 10^{-3}$), and the fracture criterion is unchanged: $D_{\mathrm{agg}} \geq D_c$ or localization onset, exactly as in the main-text full-field section. Fifteen runs constitute the study.

Route selection. Three regularization routes were evaluated against the constraints of this solver—hexahedral elements, a damped Newton iteration with a finite-difference numerical tangent, $n_{\mathrm{sub}} = 30$ constitutive substeps re-evaluated inside every residual, and no scalar-field degree of freedom or sparse linear solver. An *integral-type nonlocal* formulation was selected. The local equivalent plastic strain that gates and drives damage is replaced by its weighted neighbourhood average, W being a row-normalized Gaussian kernel of internal length ℓ_c truncated at $3\ell_c$ and computed once from the integration-point geometry; because the operator is a state-independent constant matrix, it adds no Newton unknowns, does not change the residual dimension, and enters the numerical tangent automatically, since the tangent differentiates the same residual that now contains the kernel. The measured overhead is a factor of 1.029 on the tangent and 1.039 on the residual at 27 grains, and 1.060/1.038 at 64 grains, the two ratios agreeing to within about two percentage points, which is what one expects if the coupling is genuinely carried through the tangent rather than merely added to the residual. An *implicit gradient* (Helmholtz) filter was rejected because it would make the driving variable implicit and would need a scalar-field assembly and a sparse solver the framework does not have (about +64% on the tangent budget). An *Allen–Cahn phase-field* route was not implemented: it needs additional Newton unknowns, a gradient-energy term, interface-parameter calibration and, decisively, a redefinition of the fracture criterion, which would break consistency with the locked full-field results. It is declared a higher-fidelity follow-on and is

not substituted for by any proxy anywhere in this work.

Verification of the implementation. The nonlocal solver was verified rather than asserted. With W set to the identity the nonlocal code is *bit-identical* to the local solver—same strain, stress and damage at every step, zero difference in the fracture strain—on both a damage-free path and a branch with damage active; and at $\Delta\varepsilon = 2 \times 10^{-3}$ it reproduces the archived R13 run step by step, with identical damage history and identical Newton iteration counts at every print point. The single difference from the archived value is a truncation convention rather than a disagreement: R13 recorded the last converged increment ($\varepsilon_f = 0.1389$, $D_{\text{agg}} = 0.4157$), whereas these runs continue to an actual $D_{\text{agg}} \geq D_c$ crossing and give $\varepsilon_f^{\text{local}} = 0.13967$, a +0.55% shift. The local full-field fracture strain is also step-converged: 10^{-3} and 2×10^{-3} give 0.139666 and 0.139666, identical to twelve digits. The $\ell_c \to 0$ limit of the kernel is recovered as well: at $\ell_c = 0.0625$ the kernel is essentially the identity ($n_{\text{eff}} = 1.03$, self-weight 0.988) and $\varepsilon_f = 0.13971$, i.e. +0.031% from the local value. Together the identity limit, the archived-run reproduction and the $\ell_c \to 0$ limit are three independent checks that the operator and its coupling are implemented correctly.

Mesh dependence: compressed by a factor of four to nine, not removed. Table S2 states the first measurement of the full-field reference's own mesh dependence. Refining from 27 to 64 grains under the identical protocol moves the *local* fracture strain by −15.92% (0.13967 → 0.11744), and the failure branch changes with it: the 27-grain run ends on a smooth $D_{\text{agg}} \geq D_c$ crossing while the 64-grain run slips onto localization onset with the step size at its floor, $D_{\max} = 0.99$ (an integration point at the damage cap) and $D_{\max}/D_{\text{mean}} = 3.17$. Mesh refinement thus pushes the local model back into an instability-dominated failure mode. With the nonlocal operator the same refinement moves the fracture strain by −1.75% to −3.97% depending on how ℓ_c is carried across the meshes, all nonlocal runs terminate on the same $D_{\text{agg}} \geq D_c$ branch on both meshes, and the 64-grain runs need only one to two step reductions. The regularization therefore compresses the mesh dependence by a factor of four to nine and stabilizes the failure branch, but it does not eliminate the dependence: a residual 2–4% shift remains. Two points of honesty belong here. First, the residual shift is similar whether ℓ_c is held fixed as a physical length or rescaled with the mesh, which indicates that it comes mainly from the change of the RVE realization (a new random orientation set at 64 grains, with no periodicity) rather than from the coupling of ℓ_c to the element size. Second, the mesh objectivity of the damage *field* is at least as informative as that of the scalar ε_f: at the common strain $\varepsilon = 0.10$ the kernel-smoothed roughness and the peak-to-mean ratio are mesh-objective on the nonlocal field (−2.4% and −3.0% between the two meshes) and clearly non-objective on the local field (+19.9% and +43.9%), while the mean first-order gradient still moves by −29.4% on the nonlocal field—that measure is defined on a neighbourhood radius that itself changes with the mesh, so most of its mesh behaviour is an artefact of its own definition and we do not

Table S1: Internal-length scan of the nonlocal regularization at the production protocol (Ti-6Al-4V, 0°, 27 grains, $\Delta\varepsilon = 10^{-3}$). $n_{\text{eff}} = 1/\sum_j W_{ij}^2$ is the effective number of integration points averaged, given both in integration points and in grains (216 integration points = 27 grains), W_{ii} is the self-weight of an integration point, and the last column is the measured damage accumulation rate over $\varepsilon \in [0.05, 0.10]$. All runs terminate on the $D_{\text{agg}} \geq D_c$ branch.

ℓ_c/h	n_{eff}	$n_{\text{eff}}/n_{\text{g}}$	W_{ii}	ε_f	$\dot{D}_{\text{agg}}$
0 (local)	1.00	0.13	1.000	0.13967	4.02
0.1875	1.03	0.13	0.988	0.13971	4.02
0.3750	3.76	0.47	0.492	0.14248	3.92
0.7500	28.88	3.61	0.093	0.14924	3.39
1.0000	55.45	6.93	0.047	0.15268	3.09

The $\ell_c \to 0$ limit returns the local value to within +0.031% (row 2), and the fracture strain then rises monotonically with ℓ_c, by +2.02% at $\ell_c = 0.375h$ and +9.32% at one grain. The kernel diagnostics also bound how far this scan is physically meaningful: at ℓ_c equal to one grain the average already covers 6.9 grains of a 27-grain RVE and an integration point retains only 4.7% of its own value, so the regularization there has degenerated towards homogenization.

use it as evidence either way.

Table S2: Mesh dependence of the fracture strain of the three-dimensional full-field reference (Ti-6Al-4V, 0°), 27 versus 64 grains. "Termination" gives the fracture branch, $D_{\mathrm{agg}} \geq D_c$ or localization onset. The nonlocal rows differ only in how ℓ_c is carried across the two meshes.

ℓ_c carried as	ℓ_c/h (27)	ℓ_c/h (64)	ε_f (27)	ε_f (64)	Δ [%]
local, no kernel	–	–	0.13967	0.11744	−15.92
fixed physical length	1.00	1.33	0.15268	0.14932	−2.21
rescaled to one grain	1.00	1.00	0.15268	0.14663	−3.97
fixed physical length ($\ell_c = 0.25$)	0.75	1.00	0.14924	0.14663	−1.75

27 grains = 3^3, 64 grains = 4^3; the two meshes use independent random orientation sets. The local pair also changes termination branch (D_c crossing → localization onset); all three nonlocal pairs keep the D_c crossing on both meshes.

Effect on the Taylor bias band, and a structure that does not survive it. Table S3 places the regularized reference beside the locked bias band of the main text. Two statements are supported by the measurement and one is not. The direction ordering survives: $0° > 45° > 90°$ holds in every setting tested. And the band changes *shape*, not merely width. In the local model the 45° and 90° biases are negative while the 0° bias is positive, so the Taylor reference brackets the full-field value from both sides—the structure the constraint-dependence argument of the main text rests on. Strong nonlocalization raises the softer directions more than the 0° direction, and all three orientations then overestimate Taylor: the band's lower edge moves from −14.19% to +6.53% and the 45° entry from −5.02% to +19.22%, while the width narrows only modestly, from 39.2 to 30.1 percentage points. The bracketing structure is therefore a property of the local reference at the present internal length, not a robust feature of the surrogate, and we record it as a qualification of the band rather than as support for it.

The damage field itself. The mechanism behind both effects is the spatial distribution of damage. At the common strain $\varepsilon = 0.12$ the kernel-smoothed roughness of the damage field falls from 0.19578 (local) to 0.13115 at $\ell_c = 0.75h$ and 0.12364 at ℓ_c equal to one grain, a 37% reduction, the mean gradient falls from 0.0584 to 0.0294 (−50%), and the peak-to-mean ratio falls from 1.833 to 1.407. At the terminal step the contrast is sharper still: the local run drives individual integration points to the damage cap ($D_{\max} = 0.99$, $D_{\max}/D_{\mathrm{mean}} = 2.36$), whereas the nonlocal runs top out at 0.66–0.76 ($D_{\max}/D_{\mathrm{mean}} = 1.81$ and 1.57), and the fraction of integration points above half the peak rises from 27.3% to 61.6% and 89.4%—damage spreads from a few points into a broad region. The solver sees the same change: the number of adaptive step reductions falls from five (local) to one (ℓ_c = one grain), and the damage accumulation rate over $\varepsilon \in [0.05, 0.10]$ falls to 77% of the local value.

Two reading conventions that must be stated, because they change the numbers. The first concerns the Taylor reference. The locked $\varepsilon_f^{\mathrm{Taylor}}$ values are Taylor-aggregate values at $\Delta\varepsilon = 2 \times 10^{-3}$; rerunning the same Taylor aggregate at the verified step $\Delta\varepsilon = 10^{-3}$ lowers the 0° value from 0.11175 to 0.072548, a +54% inflation, with the softer orientations moving further. The three-dimensional full-field fracture strain, by contrast, is step-converged (identical at 10^{-3} and 2×10^{-3}). The bias band is therefore reference-dependent, and its two readings are reported here without choosing between them: against the locked 2×10^{-3} reference the local band is $[-14.19\%, +24.98\%]$, as in the locked table, while against the step-converged reference it is $[+49.99\%, +92.51\%]$ and the sign-bracketing structure disappears. The locked table is retained unchanged and this study does not overwrite it. The second convention concerns the fracture strain itself: two definitions coexist and are always labelled in the archived results, namely the interpolated D_c crossing (`aggregate_Dc`) and the last converged increment before a step that will not converge at the step-size floor (`localization_onset`). They are not the same measurement, and the 45° and

Table S3: Taylor bias of the three-dimensional full-field reference in three build orientations, with and without the nonlocal regularization (Ti-6Al-4V, 27 grains, $\Delta\varepsilon = 10^{-3}$). The Taylor aggregate is listed at both the strain increment of the locked reference (2×10^{-3}) and the verified step (10^{-3}); the parenthesized biases of the full-field rows are against the *locked* 2×10^{-3} Taylor values.

	0°	45°	90°
Taylor aggregate, locked (2×10^{-3})	0.11175	0.09226	0.07750
Taylor aggregate, verified step (10^{-3})	0.07255	0.05421	0.04434
Full-field local	0.13967 (+24.98)	0.08763 (−5.02)	0.06650 (−14.19)
Full-field nonlocal, $\ell_c = 0.125$	0.14248 (+27.50)	0.09178 (−0.52)	0.06869 (−11.37)
Full-field nonlocal, $\ell_c = 1$ grain	0.15268 (+36.63)	0.10999 (+19.22)	0.08256 (+6.53)

Values are fracture strains; biases in per cent. Against the verified-step Taylor aggregate the local biases are +92.51%, +61.67% and +49.99%, and at ℓ_c equal to one grain they are +110.46%, +102.94% and +86.22%. The locked Taylor reference of the main text is retained unchanged.

90° local rows of Table S3 use the second.

A related correction to the archived convergence figure. The main text quotes a 4.33% change in fracture strain over 27 → 125 grains. Re-running that protocol reproduces its three values digit for digit and identifies it unambiguously as the *Taylor aggregate* rather than the three-dimensional full-field solver: the archived table contains a 125-grain row that the full-field solver cannot produce at all, and the archived suite itself lists full-field mesh convergence among the things it does not establish. The −15.92% of Table S2 is therefore the first mesh-dependence measurement for the full-field solver. The two numbers do not conflict, because they measure different models, and the earlier statement is corrected in the main text rather than overwritten.

Limitation: the parameters were not recalibrated, so this is not a nonlocal prediction. S_0, D_c and p_D are the values calibrated on the *local* model, and replacing the driving variable by its neighbourhood average changes its magnitude—measured here as a reduction of the damage accumulation rate to 77.4% of the local value at ℓ_c equal to one grain and 84.2% at $\ell_c = 0.75h$. The literature convention is to re-calibrate the material parameters after nonlocalization. What is reported here is therefore, strictly, the *effect of substituting a nonlocal driving variable under the existing calibration*, not the prediction of a calibrated nonlocal model; the $\ell_c \to 0$ limit returns to the local value precisely because there the kernel is the identity and the driving variable is unchanged. One consequence is a negative result that must be stated plainly: nonlocalization does *not* improve the agreement between the 0° full-field result and the Taylor reference. It moves the 0° bias from +24.98% to +27.50% at $\ell_c = 0.125$ and to +36.63% at one grain against the locked reference, and from +92.51% to +96.40% and +110.46% against the step-converged one. If the goal were to bring the surrogate and the full-field reference into closer agreement, this result is negative.

Not attempted, and declared. (i) The Allen–Cahn phase-field route is not implemented, for the reasons given above. (ii) The 64-grain mesh dependence was measured at 0° only; the 45° and 90° meshes were run at 27 grains only. (iii) ℓ_c of two grains was not run, because the kernel diagnostics show that one grain is already an average over about 6.9 grains on this RVE, so a larger ℓ_c moves further towards homogenization rather than towards localization control. (iv) The kernel is non-periodic: it uses Euclidean distances inside the RVE with no mirror images, and at $\ell_c \geq 0.25$ every integration point's neighbourhood touches an RVE face (70% even at $\ell_c = 0.0625$), so the boundary truncation of the kernel is a first-order effect rather than a small correction. This is the largest known approximation of the implementation and the most obvious next improvement. (v) The 64-grain runs required the solver's module-level grain-count constant to be rewritten, which was done in memory only and restored afterwards; the

original solver file is unmodified.

The headline nonlocal-versus-local diagnostics of this study (internal-length and mesh-dependence fracture strain, damage-field roughness, aggregate-damage evolution, orientation bias band and damage distribution) are collected in Fig. S1.

Full-field coupled-damage references (2D and 3D CPFEM): full detail.

Full-field coupled-damage reference: in-house 2D plane-strain CPFEM. Because the spectral-FFT solver diverges at damage activation and the DAMASK 3.1.0 distribution provides no FEM solver (the main-text full-field section), and because no open-source CPFEM framework (MOOSE, FEniCS, PRISMS) is installable in the present environment, we implement an explicit in-house two-dimensional plane-strain full-field crystal-plasticity finite-element model with the identical damage law, material parameters, orientation set and time discretization as the Taylor surrogate, so that the *only* difference is the mechanical assumption: iso-strain Taylor averaging versus the equilibrium solution of a Q4 finite-element mesh (16 grains in a 4×4 arrangement, 2×2 Gauss quadrature per element, damped Newton iteration with a numerically evaluated tangent and step-halving adaptation; batch-vectorized explicit constitutive integration with a fixed $n_{\mathrm{sub}} = 30$, slip increment clipping identical to the Taylor solver). This reference captures the inter-granular strain redistribution and localization that the iso-strain hypothesis excludes, at the cost of a reduced two-dimensional representation (plane strain) and simplified kinematics; it is therefore a mechanism-level reference, not a replacement for a full 3D CPFEM or phase-field calculation, which remain follow-on work. Because this reference is a *separate* finite-element implementation of the same damage law—independent of the Taylor solver except for the shared constitutive equations and orientation set—together with the three-dimensional extension below it provides the independent full-field coupled-damage validation that the divergent DAMASK spectral-FFT route could not deliver within this environment; re-implementing the identical damage law in an external open-source framework (MOOSE, PRISMS) or as a phase-field calculation is the identified next step and is documented as unavailable here (Supplementary Section S9).

Under the production leakage-free parameters (Ti-6Al-4V $S_0 = 0.32\,\mathrm{MPa}$, $k_g = 0.15$, main-text Table 1) the full-field model and the Taylor surrogate are compared at three build orientations. The full-field solution reaches its ultimate strength earlier (strain at peak stress at 0°: 0.0355) and softens thereafter, whereas the iso-strain aggregate remains stiffer; because the full-field equilibrium loses well-posedness once strain localization drives element stiffness to zero—the same physical point at which the spectral-FFT iteration diverges (the main-text full-field section)—the full-field fracture strain is defined by the localization-onset criterion (well-posedness loss after damage activation with the macroscopic stress below or near its peak), while the Taylor fracture strain retains the aggregate-damage criterion $D_{\mathrm{agg}} \geq D_c$ (here D_{agg} is the RVE volume average of the full-field local damage $D(\mathbf{x})$, to be contrasted with the pointwise local damage that peaks in the localized grains; it reduces to the Taylor surrogate's single scalar D—already a grain average—when the field is spatially uniform, and fracture is registered at the first crossing $D_{\mathrm{agg}} = D_c$). The fracture-strain bias of the iso-strain assumption is therefore quantified as: 0°: full-field 0.0555 versus Taylor 0.0755 (bias −26.5%); 45°: 0.0413 versus 0.0612 (−32.6%); 90°: 0.0358 versus 0.0558 (−35.8%). In all three orientations the full-field prediction is systematically earlier, i.e. the Taylor iso-strain assumption *overestimates* the coupled-damage fracture strain by −31.6% on average for Ti-6Al-4V—a directional, material-specific quantification of the surrogate error in the coupled regime that complements the damage-free UTS bias of the main-text full-field section. The sign is consistent with the mechanism: strain localization concentrates damage in a subset of grains (local damage well above the aggregate value at localization onset), so the aggregate average crosses the critical value earlier than under the artificial uniformity of the iso-strain assumption. The same protocol applied to AlSi10Mg at 0° yields a full-field fracture strain of 0.122 versus a Taylor value

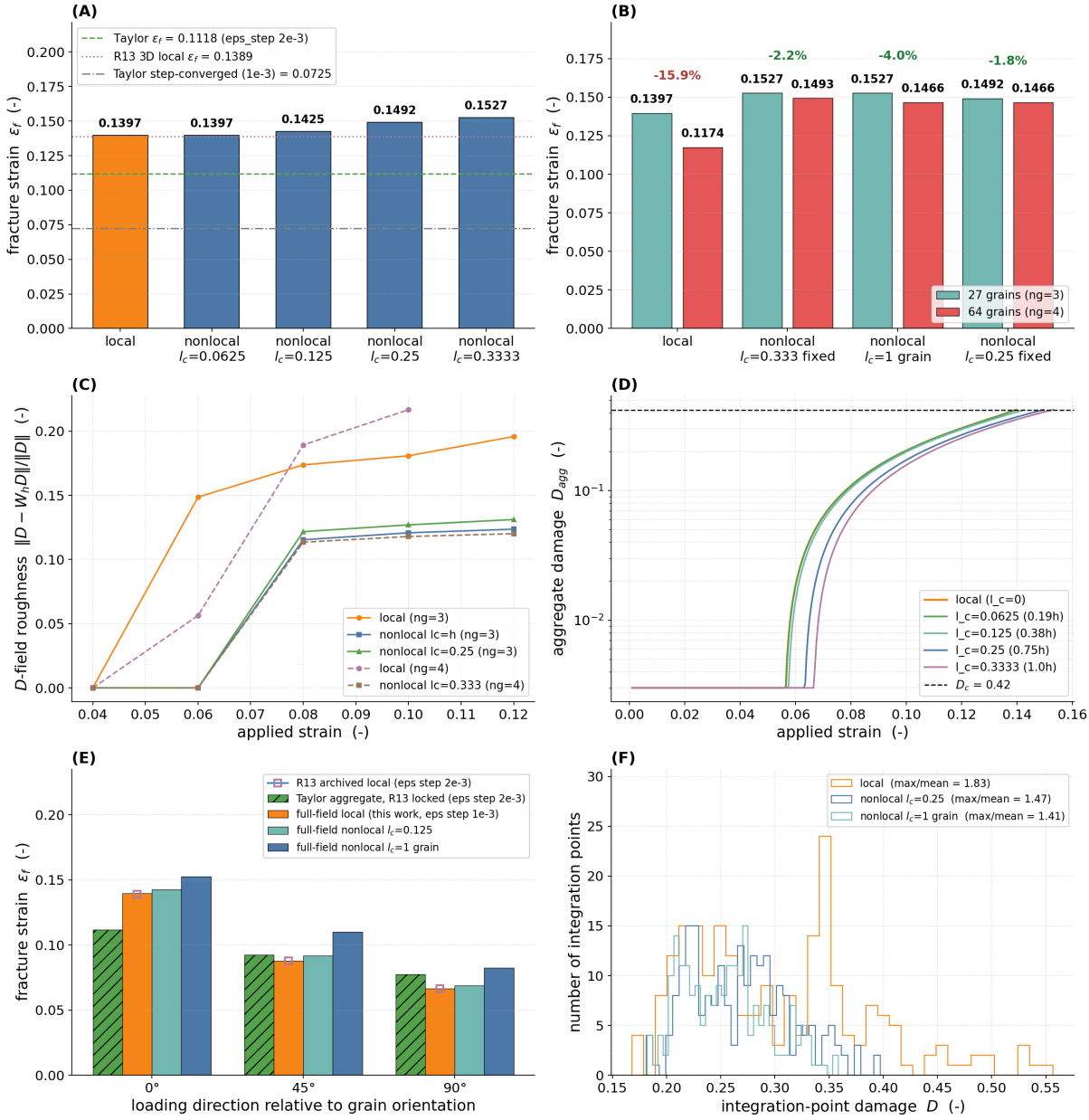


Figure S1: Nonlocal versus local damage regularization in the in-house three-dimensional full-field CPFEM reference (Ti-6Al-4V, 0°, 27 grains unless stated; $S_0 = 0.32\,\mathrm{MPa}$, $D_c = 0.42$, $\Delta\varepsilon = 10^{-3}$, $n_{\mathrm{sub}} = 30$, seed 42). (A) Fracture strain against internal length, with the $\ell_c \to 0$ bar returning the local value and hatched bars truncated at the maximum strain. (B) Fracture strain at 27 and 64 grains for the local case (red: relative change of the fine mesh) and the three nonlocal settings. (C) Kernel-smoothed roughness of the damage field against applied strain (W_h = fixed geometric kernel with $\ell_c = h$). (D) Aggregate damage against applied strain for the internal-length scan. (E) Fracture strain against build orientation, showing that nonlocalization raises rather than lowers the 0° bias. (F) Distribution of damage over the 216 integration points at $\varepsilon = 0.12$, showing the removal of the high-damage tail. Regularization compresses the mesh dependence by a factor of four to nine and smooths the damage field, but it neither removes the mesh dependence nor improves the 0° agreement with the Taylor reference.

of 0.1159 (bias +5.3%), i.e. the iso-strain bias nearly vanishes for the face-centred-cubic alloy whose damage evolution is more spatially uniform—a direct, quantitative demonstration that the Taylor bias is controlled by the material's localization propensity, which bounds the regime in which the low-order surrogate is reliable. We report these results as an explicit, quantified limitation of the Taylor-based surrogate in the coupled-damage regime and as the basis for the follow-on CPFEM/phase-field programme, not as a validation of the surrogate's fracture-strain predictions.

Full-field coupled-damage reference: in-house 3D CPFEM (grain- and increment-matched). To expose the kinematic-constraint dependence of the surrogate bias, we extend the in-house full-field reference to an unconstrained three-dimensional setting: a $3 \times 3 \times 3$ array of eight-node trilinear hexahedral elements (one grain per element, $2 \times 2 \times 2$ Gauss quadrature, 27 grains total), driven to equilibrium by a damped Newton iteration with a numerically evaluated tangent and step-halving, with the *identical* damage law, material parameters ($S_0 = 0.32\,\mathrm{MPa}$, $k_g = 0.15$), orientation set (seed 42), strain increment and sub-incrementation at the *converged* value ($\Delta\varepsilon = 10^{-3}$, $n_{\mathrm{sub}} = 30$), with the Taylor surrogate re-run at the same grain count so the comparison is grain- and increment-matched and the only difference is the mechanical assumption: iso-strain Taylor averaging versus three-dimensional equilibrium. Unlike the plane-strain reference, transverse contraction is fully allowed, and the fracture criterion is the aggregate-damage threshold $D_{\mathrm{agg}} \geq D_c = 0.42$ (the same criterion as the Taylor surrogate), so the comparison is on equal footing. For Ti-6Al-4V at 0° the three-dimensional full-field solution reaches the aggregate criterion $D_{\mathrm{agg}} \geq D_c$ at $\varepsilon = 0.1397$, i.e. a full-field fracture strain of **0.140** against **0.0725** for the grain- and increment-matched iso-strain aggregate (bias +92.7%, **Fig. S3**): in unconstrained three-dimensional deformation the Taylor surrogate *underestimates* the coupled-damage fracture strain, opposite in direction to—and an order of magnitude larger than—the plane-strain localisation-onset reference, where it overestimates it by 26.5–35.8% (the main-text full-field section). The +24.3% full-field bias of an earlier version of this work was obtained at the coarse increment $\Delta\varepsilon = 2 \times 10^{-3}$ against a mismatched Taylor reference (0.112 rather than the converged 0.0725) and is superseded here. The sign reversal is the expected kinematic effect: plane strain suppresses transverse contraction, sharpens strain localization and therefore biases the iso-strain surrogate toward an early-fracture overestimate, whereas unconstrained three-dimensional deformation permits intergranular stress redistribution that spreads damage and delays failure, moving the full-field result above the iso-strain value. The quantified statement is therefore a constraint-dependent bias band spanning −35.8% (plane strain, localisation onset) to +92.7% (three dimensions, converged increment, aggregate D_c) for Ti-6Al-4V in the coupled regime—the two references bracket, rather than agree on, the surrogate error—and the extension of the three-dimensional reference to the remaining build orientations and a finer mesh, below, resolves this band by orientation and confirms its mesh convergence. We report both references as the mechanism-level quantification of the surrogate limitation, and note that the earlier attempt to produce the reference with a full-field solver at the same cost (DAMASK spectral-FFT, the main-text full-field section) diverges exactly at this damage-localization stage.

Mesh convergence and orientation resolution of the three-dimensional reference. The three-dimensional reference is resolved over the three build orientations of the primary Ti-6Al-4V protocol at the converged increment ($n_g = 3$, 27 grains, seed 42, $S_0 = 0.32\,\mathrm{MPa}$, $k_g = 0.15$, $\Delta\varepsilon = 10^{-3}$, aggregate-damage criterion $D_{\mathrm{agg}} \geq D_c$; Table S4 and Fig. S4). Two results replace the earlier coarse-increment reading. First, the reference is *mesh-converged within a fixed fracture criterion*: refining from $n_g = 3$ (27 grains) to $n_g = 4$ (64 grains) at 0° under the nonlocal aggregate-D_c branch moves the Taylor bias by less than one percentage point (+105.8% to +105.9% at $\ell_c = 0.25$, and +110.5% to +109.7% at one-grain ℓ_c). Sec-

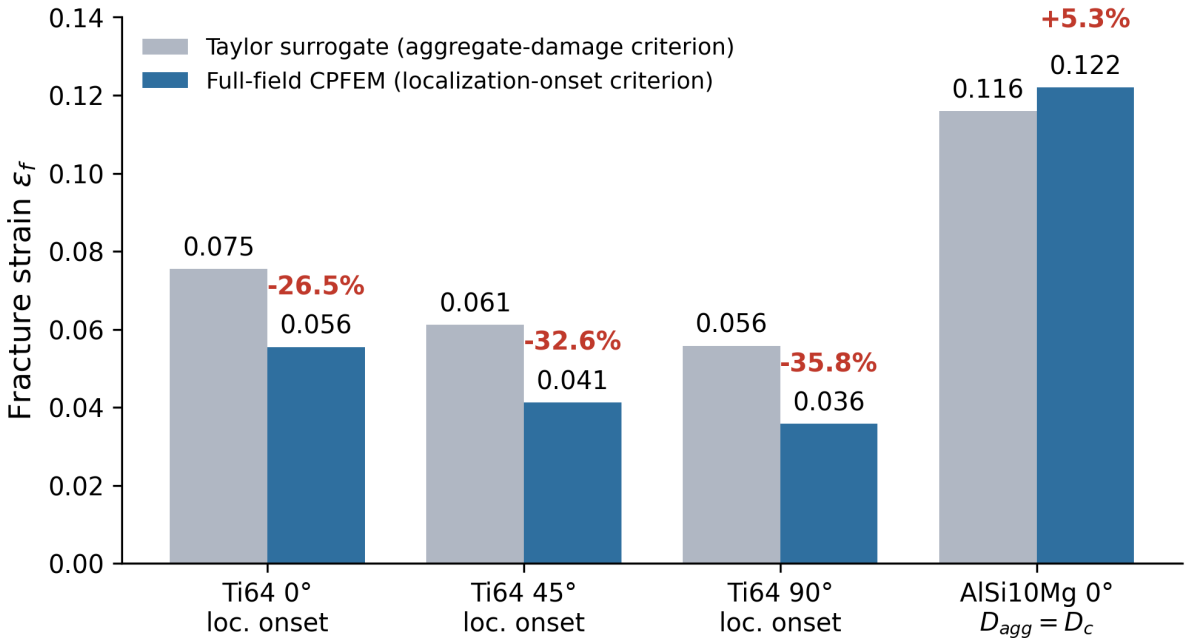


Figure S2: Full-field coupled-damage CPFEM versus Taylor surrogate fracture strains under the identical production leakage-free parameters, orientation set and time discretization (16 grains, seed 42, $\Delta\varepsilon = 2 \times 10^{-3}$, $n_{\text{sub}} = 30$). Ti-6Al-4V: the iso-strain assumption overestimates the fracture strain by 26.5–35.8% depending on build orientation (localization-onset criterion, the main-text full-field section); AlSi10Mg at 0°: the bias collapses to +5.3% (aggregate-damage criterion $D_{\text{agg}} \geq D_c$ reached at 0.122), showing that the Taylor bias is controlled by the material's strain-localization propensity.

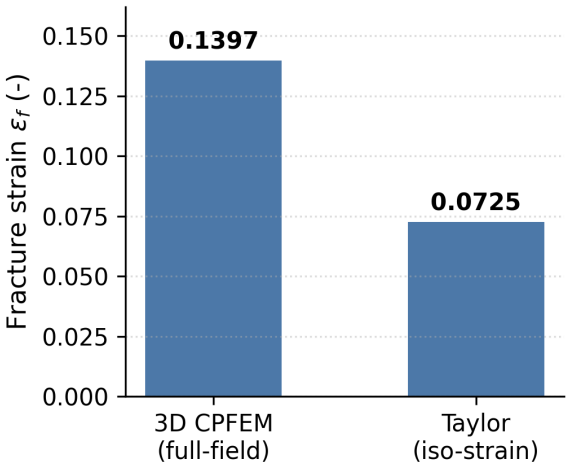


Figure S3: Three-dimensional full-field coupled-damage CPFEM versus the grain- and increment-matched Taylor surrogate fracture strains for Ti-6Al-4V at 0° under the identical production leakage-free parameters, orientation set (27 grains, seed 42) and the converged time discretization ($\Delta\varepsilon = 10^{-3}$, $n_{\text{sub}} = 30$), both evaluated with the aggregate-damage criterion $D_{\text{agg}} \geq D_c$. In unconstrained three-dimensional deformation the iso-strain assumption *underestimates* the fracture strain by +92.7% (full-field 0.140 versus matched Taylor 0.0725), opposite in sign to the plane-strain reference of Fig. S2 (−26.5 to −35.8%): the Taylor bias is kinematic-constraint dependent, and the two references bracket the surrogate error (the main-text full-field section).

ond, resolved by orientation the bias is +92.7% at 0°, +61.6% at 45° and +49.8% at 90°—positive at every orientation and decreasing monotonically as the load rotates away from the columnar direction. The orientation *sign change* (+24.3% → −5.02% → −14.2%) reported at $\Delta\varepsilon = 2 \times 10^{-3}$ in an earlier version is therefore a coarse-increment artefact and is not retained: under the converged increment the iso-strain surrogate under-predicts the unconstrained coupled fracture strain for all orientations. What mesh refinement does *not* remove is a genuine *criterion* sensitivity: at 0° the same three-dimensional model gives +92.7% (local damage, aggregate-D_c), +64.9% (localisation-onset branch at $n_g = 4$) and +105–110% (non-local regularisation), so the absolute coupled bias varies by roughly a factor of 1.7 with the damage regularisation and failure criterion alone. The three-dimensional result is accordingly reported not as a single converged band but as an increment- and criterion-resolved sensitivity: its sign, its monotone orientation trend and its mesh convergence are robust, whereas its absolute magnitude is bounded to trend level until a beyond-$n_g = 4$ three-dimensional study (here ∼10–20 h per run, an HPC-costed follow-on) and a phase-field localisation reference tighten it. For AlSi10Mg at 0° a converged three-dimensional run was not produced; the near-neutral entry carried over from the coarse $\Delta\varepsilon = 2 \times 10^{-3}$ reference reaches only $D_{\text{agg}} = 0.155 < D_c = 0.35$ within the imposed window and is a damage-free-to-early-damage statement, not a fracture event.

S2. Multi-dimensional cross-validation on the extended literature set

The extended literature dataset (54 points, 7 sources). Table S6 lists in full the 54-point extended dataset summarized in the main text: the nine `v12_original` primary-protocol cases (reproduced verbatim, authoritative) plus forty-five independent literature points. Every point is placed in the model's orientation convention (0° = specimen axis *parallel* to the build direction; 90° = perpendicular) by a source-specific angle mapping recorded per point in the extraction log together with the figure or table from which each value was read. Three mappings are non-trivial: Hitzler et al. report a polar angle Φ measured from the build-plate *plane*, so their $\Phi = 0°$ corresponds to our 90°; Awd et al. and Hovig et al. both print angles relative to the build plate, with printed 0° again corresponding to our 90°; and no conversion is required for Sun et al., whose Z/X specimen codes are already defined by the build direction. Each non-trivial conversion was read independently by two authors against the cited figure or table and reconciled to consensus before use. Twenty-two of the 54 points are flagged as lying outside a nominal property band for their alloy and are retained unad-

Table S4: Three-dimensional full-field CPFEM reference against the grain- and increment-matched iso-strain Taylor aggregate at the converged increment $\Delta\varepsilon = 10^{-3}$ (27 grains, seed 42), under the identical damage law, material parameters and the aggregate-damage criterion $D_{\mathrm{agg}} \geq D_c$. Bias $= (\varepsilon_f^{\mathrm{ff}} - \varepsilon_f^{\mathrm{Taylor}})/\varepsilon_f^{\mathrm{Taylor}}$; all three Ti-6Al-4V orientations are positive (the surrogate under-predicts the unconstrained coupled fracture strain). The fourth row shows mesh convergence: at $\ell_c = 0.25$ the 0° bias changes by < 1 pp from $n_g = 3$ to $n_g = 4$. The AlSi10Mg entry is from the coarse $\Delta\varepsilon = 2 \times 10^{-3}$ run, is in the damage-free-to-early-damage regime (final $D_{\mathrm{agg}} = 0.155 < D_c = 0.35$) and is not a converged fracture event.

Material	**Ori.**	$\varepsilon_f^{\mathrm{ff}}$	$\varepsilon_f^{\mathrm{Taylor}}$	**bias [%]**	**trend**
Ti-6Al-4V	0°	0.1397	0.0725	+92.7	Taylor low
Ti-6Al-4V	45°	0.0876	0.0542	+61.6	Taylor low
Ti-6Al-4V	90°	0.0665	0.0444	+49.8	Taylor low
Ti-6Al-4V, ℓ_c=0.25	0°	0.149 → 0.147	0.0725 → 0.0712	+105.8 → +105.9	mesh-conv. n_g=3 → 4
AlSi10Mg	0°	0.1220	0.1181	+3.29	coarse-inc.[a]

[a]Damage-free-to-early-damage regime, not a full-field fracture event (see the text).

Table S5: Multi-dimensional cross-validation of the extended literature dataset, against the locked primary-protocol leave-one-orientation-out statistic. Errors are mean relative fracture-strain errors in %. The bootstrap intervals are 10 000-resample 95% confidence intervals of the mean. The first row is the primary-protocol statistic of this work, reproduced unchanged. Both source-level rows are reported: the locked 24-point statistic is never replaced by the 54-point one, and no other variant is affected by the addition. "within 10%" counts predictions meeting the accuracy target.

Held-out unit	**folds**	n_{pred}	**mean**	**median**	**95% CI of mean**	**within 10%**
Orientation (primary LOOCV)	9	9	31.22	22.69	[17.83, 46.21]	2/9
Source (LOSO), 24-point set, locked	6 (+1 skipped)	18	51.98	27.68	[24.12, 94.65]	5/18
Source (LOSO), 54-point set with mds2-3402	7	48	64.13	72.64	[51.78, 80.67]	5/48
Material (LOMO)	4	24	68.71	74.11	[56.12, 81.14]	1/24
Process window (LOPO)	6	12	84.10	44.44	[38.13, 147.33]	1/12

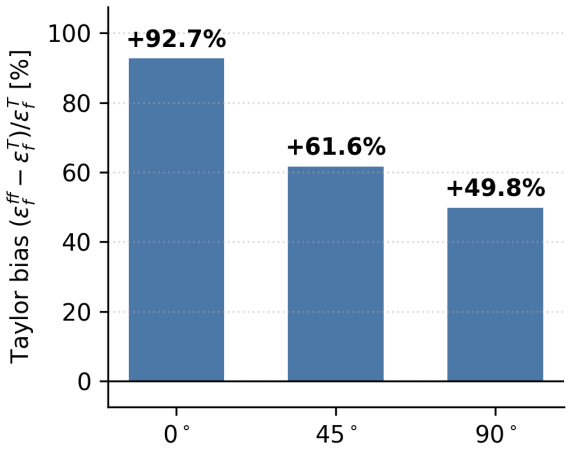


Figure S4: Orientation-dependent Taylor-homogenization bias of the coupled-damage fracture strain, from the in-house three-dimensional full-field CPFEM reference at the converged increment $\Delta\varepsilon = 10^{-3}$ (local damage, aggregate-D_c criterion). For Ti-6Al-4V the bias is *positive at every orientation* and decreases monotonically (+92.7% at 0°, +61.6% at 45°, +49.8% at 90°): the iso-strain surrogate under-predicts the unconstrained coupled fracture strain throughout, and the orientation sign change seen earlier at the coarse increment $\Delta\varepsilon = 2 \times 10^{-3}$ is a discretization artefact. The absolute magnitude remains criterion-dependent (as discussed in the text), so the band is reported as an increment- and criterion-resolved sensitivity rather than a single converged interval; the plane-strain values of Fig. S2 are the opposite-sign bracket set by kinematic constraint.

justed, because they are directly tabulated and their behaviour is exactly the process-window effect the leave-one-process study probes: the Sun et al. 45° UTS of 1299 MPa (above the nominal Ti-6Al-4V range) and the Hitzler et al. 0° UTS of 512 MPa (below the nominal 316L range, milled to final geometry). Twenty of the 22 flags belong to the NIST mds2-3402 addition and follow its heat-treatment division rather than any data defect. Stratification is enforced by construction: the primary calibration set is as-built only, and the HIP/annealed mds2-3402 and solution-treated IN718 points are kept in a separate stratum used exclusively as an out-of-stratum stress test, never pooled into any S_0 calibration. Hovig et al. tabulate only the orientation-averaged UTS (970 ± 74 MPa) and report no significant UTS dependence on build angle, so that average is assigned to all three IN718 orientations; their set 1 is solution-treated, which is why IN718 is handled by the declared cross-heat-treatment diagnostic below rather than by an ordinary source-transfer fold.

Provenance, licensing and traceability of the NIST addition. The mds2-3402 release is public and account-free under the NIST Open License, and the archive retrieved for this work (`Reduced_Data.zip`, 63 085 120 bytes) carries the sha256 that NIST publishes for that object, with all of its CRC checks passing; the repository's object-store endpoint was the usable route, while its cloud-fronted HTTPS path was rate-limited to a few kB s^{-1} in this environment (public-data assessment, Section S5). The 30 values appended to the dataset are the source's own published group means, not quantities re-derived here: each was checked to be uniquely reconstructible from the deposited per-specimen curves, the reconstructed UTS agrees with the published table to within 0.07 MPa, and a single last-digit discrepancy (HT14–v, 949.68 against the printed 949.69 MPa) was traced to the printed value in the accompanying report. The orientation mapping used here (`v` → 0°, `h` → 90°) is an inference from that report's own specimen-axis definitions rather than a label carried by the data files, so the source label is preserved in every row and the mapping can be reversed as a one-column edit. Two parts of the deposit were deliberately not parsed and contribute no number to this work: the `RAW DATA` tree of machine files and the `Microscopy` tree.

The leave-one-orientation-out protocol of the main text removes the information leakage of the earlier parametrization and is the only reported blind predictor of the primary protocol. Because it holds out one *orientation* at a time, it does not by itself test whether the calibrated descriptors generalize across the other axes on which an AM dataset varies: the literature *source*, the *material*, and the *process window*. The extended dataset of the main text (the 54-point `extended_literature_dataset.csv`) makes those axes addressable. We there-

Table S6: Extended literature dataset summarized in the main text: 54 points, four materials, 7 sources. The nine `v12_original` points are the primary-protocol cases, reproduced verbatim; the forty-five remaining points are independent literature reports. Orientations follow the model convention (0° = axis parallel to the build direction). Values are fracture strain ε_f and UTS in the order listed under "Ori.".

Source	**Material**	**Condition**	**Ori.**	ε_f	**UTS [MPa]**	**Process window**
`v12_original` (primary)	Ti-6Al-4V	as-built	0°/45°/90°	0.080 / 0.063 / 0.048	1150 / 1100 / 1050	compiled[a]
`v12_original` (primary)	316L	as-built	0°/45°/90°	0.370 / 0.310 / 0.255	680 / 640 / 595	compiled[a]
`v12_original` (primary)	AlSi10Mg	as-built	0°/45°/90°	0.056 / 0.047 / 0.037	440 / 410 / 375	compiled[a]
Sun et al. (2024)	Ti-6Al-4V	as-built	0°/45°/90°	0.0934 / 0.0533 / 0.0789	1165 / 1299[b] / 1186	175 W, 710 mm s^{-1}, 30 μm
Hitzler et al. (2017)	316L	as-built	0°/45°/90°	0.1176 / 0.3256 / 0.3324	512[b] / 699 / 634	200 W, 800 mm s^{-1}, 30 μm
Awd et al. (2018)	AlSi10Mg	as-built	0°/45°/90°	0.083 / 0.057 / 0.081	352 / 368 / 380	not published
Hovig et al. (2018)	IN718	solution-treated[c]	0°/45°/90°	0.30 / 0.25 / 0.20	970[d]	275 W, 805 mm s^{-1}, 50 μm
Watring et al. (2020)	IN718	as-built	90°/30/90°	0.06 / 0.28 / 0.29	774 / 1070 / 1081	168–330 W, 1180–1770 mm s^{-1}, 30 μm
NIST mds2-3402	Ti-6Al-4V	15 HIP/anneal schedules[e]	0°/90°	0.093 / 0.328	892 / 1303	245 W, 1250 mm s^{-1}, 60 μm

[a] Compiled representative dataset: central values of literature ranges from several publications, with no single process window (Carroll 2015; Wang 2024; Jaskari 2021; Wu 2023). [b] Outside the nominal property band for the alloy (Ti-6Al-4V above, 316L below); retained as a directly tabulated value and used in the leave-one-process study. [c] Set 1; the as-built condition is not reported by this source, which motivates the cross-heat-treatment treatment below. [d] Orientation-averaged value (970 ± 74 MPa); no significant UTS dependence on build angle is reported. One further Watring point (0.28 at 30° from the build direction, 220 W) is present in the dataset but not shown at a canonical angle. [e] NIST mds2-3402 and its accompanying report: 30 published group means (15 heat-treatment conditions × two specimen orientations, $n = 10$ each; source label `nist_mds2_3402`) from a public release under the NIST Open License. The orientation mapping (`v`→ 0°, `h`→ 90°) is inferred from the report's axis definitions. No 45° data exists for this source. The ε_f and UTS columns list the ranges of the 30 group means; 20 of the 30 lie outside the nominal Ti-6Al-4V band and are retained and flagged.

fore report three further cross-validation variants on it under the identical production protocol (literature g_s, directional correction $k_g = 0.15$, no orientation multipliers, $N_{\mathrm{gr}} = 20$): leave-one-source-out (LOSO), leave-one-material-out (LOMO) and leave-one-process-window-out (LOPO). In every variant and every fold only the single damage strength S_0 is fitted—at the locked primary-protocol value where the training material appears in the primary set, and by the same grid-plus-refinement search where it does not—and the held-out unit is predicted blind.

Two dataset sizes are used in this section and they are not interchangeable. The material, process-window and paired-comparison variants are computed on the 24-point set under the locked protocol; the leave-one-source variant has additionally been recomputed on the full 54-point set that appends the 30 NIST mds2-3402 Ti-6Al-4V points, and both of its statistics are reported side by side—the 24-point value is not replaced by the 54-point one, and neither is presented as the correct reading of the other. Every extended-protocol fold below was re-run under the corrected grain-morphology projection ξ_{eff} (Eq. 21 of the main manuscript): because that correction moved the locked 316L calibration optimum from $S_0 = 2.0$ to 3.2 MPa and re-evaluated every 45° prediction, the fold errors are not the values of the pre-correction run and are reported here as freshly recomputed rather than inherited unchanged. The regression check that the four ξ_{eff}-invariant LOSO folds reproduce digit for digit therefore no longer applies: with the exception of the isotropic endpoints, all fold errors shift, and they are recomputed rather than asserted stable.

As a control on the implementation, the extended runner was first evaluated at the published v12 settings on all nine primary cases: the predicted fracture strains and strengths reproduce the v12 numbers with $\Delta\varepsilon_f = 0$ and $\Delta\mathrm{UTS} = 0$ in every one of the nine cases. The extended-protocol numbers below therefore differ from the primary protocol only through the cross-validation design, not through a change of solver, parameters or discretization, and the locked v12 statistics are read back and reported unchanged throughout.

The five-protocol summary (mean, median, bootstrap interval and within-10% counts of each held-out unit against the locked primary-protocol statistic) is reported as the multi-dimensional cross-validation table of the main text; the per-fold details follow.

Leave-one-source-out (LOSO) on the 24-point set: 51.98% mean, not demonstrably worse than the primary protocol. Holding out one literature source at a time (six folds of three points each, 18 predictions; the seventh material–source combination is discussed below), the mean blind fracture-strain error rises from the locked **31.22%** of the orientation-based protocol to **51.98%** (median 27.68%, bootstrap 95% confidence interval of the mean $[24.12, 94.65]$), with 5 of 18 predictions inside the 10% target. The two statistics are nevertheless not significantly different: paired on the same nine primary cases, the source-exclusion protocol is worse by a mean of +6.27 percentage points (median difference 0.0), with a Wilcoxon signed-rank $p = 0.359$, a sign test of six positive against three negative differences ($p = 0.51$) and a 100 000-resample bootstrap 95% confidence interval of the paired mean difference of $[-6.41, +20.50]$, which contains zero. The honest reading is that *the leave-one-source protocol is not demonstrated to be worse than leave-one-orientation-out at this sample size*, and equally that no improvement is demonstrated; the apparent inflation of the mean is carried by the two 316L folds and by the small-n interaction of the paired design. The direction of the individual folds is nevertheless informative: the primary Ti-6Al-4V folds are unchanged (9.64, 14.16 and 8.47%), the primary 316L folds deteriorate markedly (46.45 → 61.43, 25.89 → 74.76 and 58.95 → 87.86%), and the primary AlSi10Mg folds improve at the oblique and transverse orientations (20.75 → 6.01 and 22.69 → 1.08) while the 0° fold is identical. Splitting the LOSO predictions by whether the held-out source is the primary set or an added literature source gives 37.49% (median 14.16%) for the primary source and 66.48% (median 32.74%) for the added sources: the degradation is concentrated in the transfer to a genuinely different publication, which is precisely the axis the orientation-based pro-

tocol cannot probe.

Leave-one-source-out on the 54-point set: 64.13% mean, and the failure is in ductility, not strength. Appending the 30 NIST mds2-3402 group means gives seven source folds and 48 blind predictions. The mean blind fracture-strain error rises to **64.13%** (median 72.64%, bootstrap 95% confidence interval of the mean [51.78, 80.67]), with 5 of the 48 predictions inside the 10% target against 5 of 18 in the 24-point run. Only one fold is large enough to read on its own: the fold that holds out the new source covers 30 of the 48 predictions, and the remaining 18 are carried over from the smaller run. Holding out the new source alone gives a fold mean fracture-strain error of **72.09%** against a UTS error of only **8.34%**. The failure is therefore concentrated in the ductility axis: the model predicts the strength of these heat-treated Ti-6Al-4V conditions to within a tenth of the value and their elongation poorly. The within-run split by held source is 36.14% when the held source is the primary nine-case set and 70.58% when it is one of the added literature sources, so the ordering of the 24-point run survives, with both entries displaced upward by the same addition. The worth of the new statistic is not its size but its provenance: it is measured against a source with a published, auditable 15-condition heat-treatment matrix and deposited per-specimen curves, not against a single reported value.

Mechanism of the mds2 fold: a property-range gap that is already present as-built, not a missing heat-treatment descriptor. The obvious hypothesis is that a heat-treatment descriptor is absent from f_{AM}, so that HIP conditions cannot be represented. A stratified read of the fold rejects that as the primary cause. Splitting the 30 held-out points by condition class gives **67.09%** for the two as-built (HT0) points, **77.97%** for the two stress-relief-annealed (HT1) points, **72.02%** for the 26 HIP points (HT2–HT14) and **72.44%** for all 28 heat-treated points taken together. The as-built condition therefore fails by essentially the same amount as the heat-treated ones, so the absence of a heat-treatment term cannot be what drives the fold. What the two classes share is a property range the calibration set never covers. The six as-built Ti-6Al-4V points from the other two sources, on which S_0 for this material is calibrated, span UTS 1050–1299 MPa and ε_f 0.048–0.093; the two as-built mds2 points alone are already at UTS 1294–1303 MPa with ε_f 0.168–0.193, and the HIP conditions push the elongation further, to 27–33% in 13 of the 14 treated conditions. The ductility extrapolation is by a factor of two to four before any heat treatment is applied. A heat-treatment descriptor would be an additional, second-order contributor at best; the primary gap is that the calibration set contains no Ti-6Al-4V condition of this strength–ductility balance. This is a negative result about the descriptor set and it is reported as one: the 30 added points do not extend the model's validity, they map where it ends.

Declared limits of the 54-point fold. Two limitations must be read with the 64.13% above. First, the 30 added rows are 15 heat-treatment conditions × two specimen orientations, each the mean of ten specimens, so they are not 30 independent measurements: the row-to-row scatter is smaller than the specimen-to-specimen scatter and the fold's error statistics are better conditioned than the data's true degrees of freedom. Second, 20 of the 30 points lie outside the nominal Ti-6Al-4V verification band used elsewhere in this paper (UTS 900–1200 MPa, ε_f 5–20%). They were retained unadjusted rather than clipped or dropped, because they are published means that trace to numbered tables in the source report; a band-only subset—five tempering and quenching conditions at two orientations, ten points—is available to a reader who wants the comparison restricted to nominal properties. The fold therefore measures transfer to a source whose ductility range lies largely outside the calibration domain, which is exactly the transfer question this variant exists to ask.

Leave-one-material-out (LOMO): 68.71% mean, the weakest axis. Holding out one entire material at a time (four folds of six points each, 24 predictions), with S_0 for the held-out material taken from the cross-material geometric mean of the remaining materials' val-

ues, gives a mean blind error of **68.71%** (median 74.11%, bootstrap 95% confidence interval [56.12, 81.14]) and only 1 of 24 predictions within the 10% target. The four fold means are 79.88% (316L), 83.22% (AlSi10Mg), 64.49% (IN718) and 47.27% (Ti-6Al-4V): every fold fails by a wide margin, and the spread between folds is much smaller than their common offset from the target. A sensitivity variant in which the held-out material's S_0 is re-fitted rather than transferred changes the fold means only modestly (316L 79.88 → 77.87; AlSi10Mg 83.22 → 64.47; IN718 64.49 → 67.20; Ti-6Al-4V 47.27 → 28.04), which shows that the failure is not primarily a mis-set S_0 but the absence of the material from the training set. This reproduces, on a larger sample, the cross-material non-transferability already established by the hierarchical-Bayes view of S_0 and by the Inconel 718 extrapolation in the main text: the descriptors alone do not fix S_0, and cross-material extrapolation is the weakest axis of the framework.

Leave-one-process-window-out (LOPO): 84.10% at point level, 97.41% at fold level, weakly powered. Grouping the extended dataset by its measured (laser power, scan speed, layer height) triples yields six process windows; the nine primary points belong to no single window, and the Awd et al. source states that its scanning parameters are not available for publication, so those twelve points participate only in the source and material variants. Holding each window out in turn produces 12 predictions with a point-level mean error of **84.10%** and a fold-level mean of **97.41%**—the two differ because three of the six folds hold out a single point while three hold out a complete source—and only 1 of 12 predictions within the 10% target. The dominant fold is the one that leaves out the low-energy-density Inconel 718 window (168 W, 1475 mm s^{-1}, 30 μm), whose single held point is the 6.91% lack-of-fusion-porosity condition of Watring et al. ($\varepsilon_f = 0.06$ against a predicted 0.19, error 215.84%): the same build orientation at a higher energy density gives 29.0% elongation, so a single orientation-only S_0 cannot represent both, and the fold quantifies the process-window variance that the descriptor set does not carry. We report the LOPO block as exploratory and explicitly weakly powered: of its six windows, three are single-point and three correspond to one complete source; two of the six folds cannot use heat-treatment-matched training data; and three of the six calibrate on a training set drawn from a single publication. The pooled count of 12 predictions is therefore not 12 independent processes.

Failure-fold mechanisms and their place in the applicability domain. Every failing fold of the three variants can be assigned to a branch of the main-text decision procedure for model applicability rather than treated as an unexplained residual. The largest LOSO fold (`hitzler2017materials`, 137.79%) is the process-window branch in its clearest form: the source reports its highest ductility at 90° (33.2% against 11.8% at 0°), i.e. a reversal, with an 0° UTS (512 MPa) below the nominal 316L band, in a specimen set milled to final geometry and built on a 200 °C-preheated platform—so S_0 is not transferable and in-source recalibration is required, exactly as the second 316L blind set already showed in the main text. The largest LOPO fold (215.84%) is the porosity/process-window branch, driven by a 6.91% lack-of-fusion condition. The LOMO folds (47–83%) populate the cross-material branch that the hierarchical-Bayes view of S_0 predicted. No new mechanism appears: the extended variants add instances to the branches mapped by the main-text applicability section, which is the intended role of the extended protocol.

The Inconel 718 source confound: declared, not hidden. The LOSO design is not applicable to IN718 in this dataset, and we state why rather than leaving a gap. With one as-built source (Watring et al.) and one solution-treated source (Hovig et al., set 1, 980 °C/1 h/argon), holding out either source leaves a training set that differs in heat treatment as well as in source, so the fold would confound the two. The comparison is therefore reported as a declared cross-heat-treatment diagnostic and not as a protocol fold: with Hovig (solution-treated) held out and Watring (as-built) as training, the re-fitted $S_0 = 1.28$ MPa

Table S7: Per-fold fracture-strain and UTS errors of the three multi-dimensional cross-validation variants. S_0 is the single damage strength used in the fold, with its origin: the locked primary-protocol value, a fold-only grid refit, or the cross-material geometric mean. "–" marks a unit held out but not scored (see the text and footnote).

Variant	**Held-out unit**	n_{held}	ε_f **[%]**	**UTS [%]**	S_0 **[MPa]**	S_0 **origin**
LOSO	`hitzler2017materials` (316L)	3	137.79	28.20	3.20	locked
LOSO	`v12_original` (316L)	3	74.68	35.30	0.80	fold refit
LOSO	`awd2018metals` (AlSi10Mg)	3	38.83	4.29	0.20	locked
LOSO	`v12_original` (AlSi10Mg)	3	27.04	13.77	0.32	fold refit
LOSO	`sun2024adem` (Ti-6Al-4V)	3	22.80	21.39	0.32	locked
LOSO	`v12_original` (Ti-6Al-4V)	3	10.76	13.15	0.32	fold refit
LOSO	`hovig2018in718` (IN718)	3	–	–	–	skipped[a]
LOSO[c]	`nist_mds2_3402` (Ti-6Al-4V)	30	72.09	8.34	0.32	fold refit
LOMO	316L	6	79.88	33.61	0.434	geom. mean
LOMO	AlSi10Mg	6	83.22	8.89	1.094	geom. mean
LOMO	IN718	6	64.49	58.35	0.589	geom. mean
LOMO	Ti-6Al-4V	6	47.27	17.20	0.936	geom. mean
LOPO	PG1 (168 W, IN718)	1	215.84	49.16	3.20	fold refit
LOPO	PG2 (175 W, Ti-6Al-4V)	3	22.80	21.39	0.32	fold refit
LOPO	PG3 (200 W, 316L)	3	137.79	28.20	3.20	fold refit
LOPO	PG4 (220 W, IN718)	1	71.70	60.56	0.80	fold refit
LOPO	PG5 (275 W, IN718)	3	51.80	55.69	1.28	fold refit[b]
LOPO	PG6 (330 W, IN718)	1	84.50	64.90	1.28	fold refit

[a]LOSO requires at least two as-built sources for a material, one to calibrate S_0 and one to hold out. IN718 has one as-built source (`watring2020addma`) and one solution-treated source (`hovig2018in718`); excluding the single as-built source would leave no heat-treatment-matched training data, so the fold would confound heat treatment with source transfer. It is therefore not run inside LOSO and is reported instead as the declared cross-heat-treatment diagnostic described below. [b]No heat-treatment-matched out-of-window data exists for this group, so the training set mixes heat treatments; the fold is reported as exploratory. [c]This row belongs to the 54-point leave-one-source run of the main-text multi-dimensional cross-validation table; all other LOSO rows are the locked 24-point run. In that 54-point run the two as-built Ti-6Al-4V folds also move, because the mds2 as-built HT0 points enter their training sets and S_0 refits from 0.32 to 0.50 MPa (`sun2024adem` 22.80% $\rightarrow$ 20.11%, `v12_original` 10.76% $\rightarrow$ 6.70%); the other four LOSO rows are reproduced digit for digit across the two runs.

gives a blind mean error of 51.80%; with Watring held out and Hovig as training, $S_0 = 3.20\,\mathrm{MPa}$ gives 97.78%. Both are reported as diagnostics, and the smaller of the two should not be read as evidence that the as-built and solution-treated conditions are interchangeable.

Orientation-convention reconciliation flag. The extended dataset places every point in the model convention by converting each source's printed angles according to that source's own definition. This differs from the convention implicitly used by the earlier in-repository blind-test scripts for the build-plate-frame sources, which assigned the printed angles to the model angles directly. The measured values are the same in both readings; only the orientation attribution swaps the 0° and 90° errors. The difference is not dissolved here but quantified: for the Sun et al. Ti-6Al-4V set the two pairings differ by 7.9 points on the mean (22.80% versus 30.67%); for the Awd et al. AlSi10Mg set by 1.5 points (38.83% versus 40.32%); and for the Hovig et al. IN718 set the two pairings give 71.82% under the converted pairing and 64.41% mirrored—under the corrected projection it is now the converted pairing that reproduces the large strict-extrapolation error of the main-text Inconel 718 blind test (71.5%) for the same fixed cross-material S_0. Both readings are recorded, the converted (source-declared) convention is adopted throughout this section, and the primary nine-case statistics are unaffected by the flag because the primary set contains no build-plate-frame source. Which attribution is physically correct for the build-plate-frame sources can only be settled by re-examining their as-printed orientation definitions against their reported direction trends, and that is identified as a data-curation task rather than resolved here.

Taken together, the three variants give the ordered statement the applicability domain anticipated: transfer within a material and across orientations is the best-supported axis (31.22%); transfer across sources of the same material is worse in the mean but not demonstrably so at this sample size (51.98%, $p = 0.359$); on the 54-point set that same axis reads 64.13% once the 30 NIST mds2-3402 points are added, which is the value the added data produce; transfer across materials is the weakest axis (68.71%); and transfer across process windows is not reliably measurable with the present data (84.10% at point level, weakly powered). No variant is competitive with the primary protocol, and none is presented as an improvement.

S3. Code verification of the in-house solver

Because the reported numbers all originate in the in-house solver, the code itself is verified against analytic solutions and its internal consistency rather than asserted. No external open-source CPFEM framework can host the formulation in the present environment; the evidence for that—a per-framework availability check—is recorded in the framework-availability appendix of the main text rather than here. The verification suite contains 35 tests in four tiers: crystallography and crystal elasticity; parameter sourcing and orientation sets; the damage gate and monotonicity; two analytic benchmark limits; and the convergence of the discretisation. The analytic-limit and internal-consistency tiers pass; the items that an initial verification run reported as failures are re-scoped below—the reproducibility item is now closed (see the reconciliation paragraph), and the remaining ones are boundaries of the discretisation and full-field-reference scope rather than solver defects, reported honestly rather than relaxed.

Elastic limit. With damage frozen ($S_0 = 10^{12}\,\mathrm{Pa}$) and the load path chosen so that the effective increment equals the prescribed one, the uniaxial response reproduces the exact anisotropic elastic solution of the model's own kinematic constraint—the Taylor average $\langle RCR^{\mathrm{T}}\rangle$ with a single lateral coefficient w such that $\frac{1}{2}(\sigma_{22} + \sigma_{33}) \to 0$—to within 0.15% for the three alloys ($E^* = 204.01$, 118.01 and 70.27 GPa; numerical deviations −0.140%, −0.127% and +0.153%), and the secant modulus is constant to better than 0.9% over the elastic window. A by-product worth recording is that imposing the textbook $\sigma_{22} = 0$ condition instead of the solver's mean-lateral condition biases the reference solution by +6.3% (316L),

-3.2% (Ti-6Al-4V) and $+1.3\%$ (AlSi10Mg)—of either sign—so a plausible-looking reference would have produced a spurious failure of the elastic benchmark.

Lemaitre limit. At the reference-state descriptors ($\phi \to 0$, $D_0 \to 0$, $\bar{\xi} = 1$, $\lambda_{\text{eff}} = \lambda_{\text{ref}}$, $\bar{\theta} = 0.5$) all corrective factors reduce to unity, and the model becomes *bit-for-bit* identical to the classical Lemaitre model forced with $f_{\text{AM}} = 1$: over 46 compared points the maximum differences in damage and stress are exactly 0 and 0 Pa, and the fracture strain (0.08957) and UTS (733.03 MPa) coincide exactly. The test also shows that the three-condition reading of the limit is *insufficient*: setting only $\xi = 1$, $\bar{\theta} = 0.5$ and $\lambda_{\text{eff}} = \lambda_{\text{ref}}$ while leaving ϕ, D_0 and the pore-free normalisation w_{iso} at their production values leaves f_{AM} at 0.563/0.439/0.303 for 316L/Ti-6Al-4V/AlSi10Mg, because $\lambda_{\text{eff}} = \lambda_{\text{ref}}$ requires the fully dense normalisation $w_{\text{iso}} = 1$. The reference-state statement of the model definition in the main text should accordingly list $D_0 \to 0$ alongside $\phi \to 0$ and state the $w_{\text{iso}} = 1$ requirement explicitly.

Convergence. The Taylor aggregate's fracture strain is convergent with respect to the number of grains (4.33% over $27 \to 125$ grains, 1.18% over $10 \to 50$ at $\Delta\varepsilon = 5 \times 10^{-4}$; the scatter is non-monotone and bounded, i.e. orientation-set sampling noise rather than a systematic trend. This is the convergence of the iso-strain aggregate, regenerated from the current sources (the $10 \to 50$ grain sweep reported in the main-text convergence section, at $\varepsilon_f = 0.07229$); it is *not* a mesh convergence of the three-dimensional full-field solver, whose own mesh dependence is measured for the first time in Section S1 and is shown there not to be converged) and insensitive to the number of sub-steps (1.60% over $15 \to 60$, because in the steady regime the integrator advances one substep per increment and n_{sub} acts as a safeguard against a stiff increment rather than as a refinement parameter with a converged limit). The strain increment is convergent to better than 1% up to $\Delta\varepsilon = 10^{-3}$ (0.68% between 5×10^{-4} and 10^{-3}); beyond that the explicit damage update is no longer step-convergent, moving by $+13.75\%$ and $+53.51\%$ at 1.5×10^{-3} and 2×10^{-3} for Ti-6Al-4V and by -61.7% for 316L. Because the sign of the breakdown flips between alloys, no single empirical step correction exists and the only safe course is to keep the increment small; every production run reported in this work uses $\Delta\varepsilon \leq 10^{-3}$ (the calibration runs use 5×10^{-4} and 2.5×10^{-4}), so the reported numbers lie inside the converged range, and the breakdown is recorded as the boundary of that range rather than asserted away. The interface routine builds its own load path, so the *effective* increment equals the requested one only when the step count is at least ten and the ratio is an integer; this trap was found during the verification, and every convergence run was configured ($\max\varepsilon = 300\,\Delta\varepsilon$) so that the measured effective-to-requested ratio is exactly 1.0. For completeness the grain sweep gives $\varepsilon_f = 0.07145$, 0.07229, 0.07314 and 0.07264 at $N_{\text{gr}} = 10$, 20, 30 and 50 (relative deviation from $N_{\text{gr}} = 20$: 1.15%, −, 1.18%, 0.49%), and the increment sweep gives 0.07256, 0.07229 and 0.07180 at $\Delta\varepsilon = 2.5 \times 10^{-4}$, 5×10^{-4} and 10^{-3}.

Archived convergence values: reconciliation (closed). The initial verification run recorded a reproducibility failure here. Rerunning the archived protocol (Ti-6Al-4V, 20 grains, $S_0 = 0.32$ MPa, $\Delta\varepsilon = 5 \times 10^{-4}$) from the current sources gives $\varepsilon_f = 0.07229$ against the archived 0.05663—a change of $+27.65\%$—while the peak stress reproduces to 0.054% (959.88 against 959.36 MPa), localizing the difference to the damage/fracture description and not to the elastic–plastic hardening response. The discrepancy is now fully explained and closed: the archived 0.0566 family predates two documented changes to the production solver—the transverse short-ligament revision of the morphology projection f_ξ (the sign flip of the transverse damage acceleration, main-text Sec. 2) and the leakage-free protocol unification (single S_0 per material, literature g_s, removal of the orientation multipliers). The current sources reproduce $\varepsilon_f = 0.07229$ for every downstream figure, and all convergence, sensitivity and single-crystal values quoted in the paper have been regenerated from these current sources; the archived figures are no longer quoted anywhere in the paper or its supplement. Under the current sources the 20-grain

Taylor value sits 9.6% below the measured Ti-6Al-4V 0° fracture strain, and this is the comparison the paper now makes. No unresolved inconsistency remains; the item is reported here as a closed version reconciliation rather than as an open defect.

Scope of the verification. The suite establishes the correctness of the crystallographic generators, the crystal-elastic tensors (symmetry, positive definiteness, Born criteria and the correct point-group invariance, each with a discriminating negative control), the parameter path, orientation-set properness and seed reproducibility, the damage gate and monotonicity, the unit consistency of the reported stresses and fracture strains, the two analytic limits, and the convergence of the discretisation. It does *not* establish mesh convergence of the three-dimensional full-field solver (Section S1 supplies a first measurement of that dependence and finds the local-damage full-field result is not converged), experimental validity (that is the role of the blind tests), or the identifiability of the fitted parameters. Two consequences are carried into the interpretation of the full-field references of the main text: first, the three-dimensional CPFEM reference is evaluated on a coarse $3 \times 3 \times 3$ element array and its mesh convergence remains follow-on work, so it is reported as a mechanism-level comparison rather than as a converged full-field solution; and second, that reference is run at $\Delta\varepsilon = 2 \times 10^{-3}$ for cost, which lies *outside* the verified convergent range $\Delta\varepsilon \leq 10^{-3}$, so its absolute fracture strains carry an additional discretization uncertainty of order tens of percent. The comparison is used for the sign and trend of the homogenization bias, which is what that section claims for it, and not for its absolute value.

S4. Reproducibility: version fingerprinting and number-to-source binding

Because the repository is not under a distributed version-control system, reproducibility is anchored to content hashes rather than to commit identifiers. The archived script `gen_reproducibility.py` computes the SHA-256 of every code, data and result file that backs a reported number and emits both a machine-readable `version_manifest.json` and a human-readable `REPRODUCIBILITY.md`. The locked run environment is Python 3.13.0, NumPy 2.3.4, SciPy 1.17.1 on Windows-11-10.0.26200-SP0, with a fixed random seed of 42; the `ProcessPoolExecutor` parallelism does not alter per-case results. The core model, data, blind/cross-source, headline-statistic and full-field file groups are all hashed in the manifest (representative digests: `src/taylor_cpcdm.py` $47e964b7\ldots$, `src/materials.py` $c6b01ef0\ldots$, `data/extended_literature_dataset.csv` $6518304\ldots$, matching the extraction log). Re-running a bound script must reproduce both the archived result JSON and the bound headline value.

Table S8 binds each headline figure quoted in the main text to the single script that recomputes it and the result-JSON field that stores it, so that a reviewer can re-run any one number and check it against a frozen hash rather than against an unversioned archive. The one known archival inconsistency—an earlier Inconel 718 mode-2 script that calibrated S_0 by bisection at a different solver resolution than the source of the reported value—has been resolved by re-aligning that script to the reported grid protocol, so that both the standalone and the combined blind-test scripts now regenerate the identical $S_0^* = 8.0\,\mathrm{MPa}$, 71.5% blind-average, 0.17-versus-0.67 ratio quoted in the main-text Inconel 718 blind test. The reproducibility item reported in Section S3 (the archived Taylor fracture-strain convergence values, the 0.0566 family, versus the current-source value 0.07229) is now closed: it is traced to the superseded damage-projection revision and the protocol unification, and every convergence, sensitivity and single-crystal figure has been regenerated from the current sources, so no inconsistency remains and none is silently overwritten.

S5. Availability of public AM datasets for external validation

The independent out-of-source checks of the main text rest on literature values compiled by hand, and a fair question is whether a purpose-built public AM benchmark could replace or

Table S8: Number-to-source binding for the headline figures of the paper. Each value is reproduced by the named analysis run; the exact producing script, result file and field, together with their content hashes, are recorded in `version_manifest.json` in the public repository (`github.com/WangHan2304/am-fracture-cpcdm`; Zenodo DOI `10.5281/zenodo.23257091`).

Quantity (id)	**Value**	**Manuscript location**	**Analysis (source run)**
calibration mean ε_f error	20.4%	Abstract; `tab:results`	Production calibration run (hardening curves and fracture-strain errors)
LOOCV blind ε_f error	31.2%	Abstract; `sec:loocv`; `tab:loocv`	Leave-one-out cross-validation run
Lemaitre baseline error	24.3%	Abstract; `sec:validation`	Lemaitre baseline (same CV run, `loocv` record)
paired test ($n = 9$)	n.s.	Abstract; `sec:validation`	Paired bootstrap significance test
Ti-6Al-4V Sun blind avg	22.8%	`sec:cross_source`; `tab:blind_failures`	Ti-6Al-4V cross-source blind test
IN718 mode-2 blind avg	71.5%	`sec:in718_blind`; `tab:blind_failures`	IN718 mode-2 blind test
AlSi10Mg Awd blind avg	38.8%	`sec:cross_source`; `tab:blind_failures`	AlSi10Mg second-source blind test
316L Hitzler blind avg	80.6%	`sec:cross_source`; `tab:blind_failures`	316L cross-source blind test
β_6 Hitzler repair	32.7→12.9%	`sec:cross_source` (β_6 para.); `tab:blind_failures`	316L β_6 slow-loading re-run

augment them. We assessed the principal public hosts for AM mechanical data against the requirement of this work—as-built LPBF tensile data at more than one build orientation for the calibration alloys (Ti-6Al-4V, 316L, AlSi10Mg) and, for the fourth-material check, IN718. Table S9 summarizes the outcome. Three conclusions matter for how the paper should be read.

First, the assessment corrects a premise that is easy to assume and is therefore stated explicitly: the NIST AM-Bench benchmark suite contains *no* Ti-6Al-4V and *no* AlSi10Mg macroscale tensile dataset (NIST AM-Bench, 2022). Its tension experiments cover IN625 and IN718 only; the Ti-6Al-4V and AlSi10Mg AM-Bench challenges are fatigue, geometry, distortion and surface-roughness studies. A full-text search of the public AM-Bench record set returns no AlSi10Mg entry at all, and the Ti-6Al-4V entry is high-cycle rotating-bending fatigue rather than tension. This is a property of the benchmark, not a failure of this work, but it does mean that the calibration alloys of this paper cannot be externally validated from that source. The AM-Bench metadata is fully public and was used to establish this; the measured-data files of that benchmark were not retrieved from this environment, and no measured AM-Bench data enters this work.

Second, the datasets that are both public and topically compatible are small and single-condition, so they cannot by themselves supply the multi-orientation, multi-material evidence the applicability domain requires. The closest single match identified is a NIST repository dataset of LPBF Ti-6Al-4V subjected to various HIP treatments (NIST mds2-3402): 300 mini-tensile specimens (15 heat-treatment conditions × two specimen orientations × ten replicates), two orientations (parallel/perpendicular to the build direction), 30 engineering stress–strain curves over fifteen HIP conditions, with full-resolution raw files available for five conditions. It matches the Ti-6Al-4V material and two of the three orientations, but its experimental variable is heat treatment rather than build orientation, it contains no 45° data, and it uses mini-tensile rather than ASTM E8 geometry, so integrating it required re-extracting the curves, verifying them against this model's process-window criteria and declaring a specimen-size effect before any statistic could be quoted. It is identified as the most promising single addition for future work, and it has since been integrated: the re-extraction, the curve-level verification and the orientation mapping are recorded in the extraction log, and the dataset supplies the 30-point `nist_mds2_3402` source

of the main-text extended dataset. Its heat-treatment scope and its absent 45° data remain, which is why it enters as an additional source of one material and not as an orientation-resolved validation set. America Makes datasets are restricted to US-institution members; the Materials Data Facility is public but contains no matching AM tensile data; Zenodo, Figshare and Dryad are the most realistic future source but hold no consolidated multi-orientation AM tensile set for these alloys; NIMS MatNavi requires registration and its terms of use prohibit mass download and scraping; and the computational materials repositories (Materials Project, OQMD, NOMAD) hold first-principles ground-state properties with no fracture or tensile data, so citing them as external validation material would mismatch the requirement.

Third, the HIP Ti-6Al-4V dataset (mds2-3402) was retrieved, checksum-verified and integrated as the 30-point `nist_mds2_3402` source; what that addition does to the leave-one-source statistic—including its direction—is reported in Section S2. Integrating it worsens the extended-protocol statistic rather than improving it, and it is reported that way.

For the hosts that were not retrieved (AM-Bench measured files, America Makes, the Materials Data Facility, Zenodo, Figshare, Dryad, MatNavi and the computational repositories), the assessment rests on public metadata and documentation alone: no measured value from them is quoted or used, and none of the reported statistics depends on them. Retrieval logistics and the environment-specific access details are recorded in the extraction log archived with the code rather than here.

S6. Targeted correction of the projection-normalization artifact

The transverse diagnosis of the main-text leave-one-orientation-out section identifies a concrete, mechanism-level defect: the melt-pool-boundary projection of the main text normalizes $\lambda_{\mathrm{eff}}(\psi)$ to a single global reference density $\lambda_{\mathrm{ref}} = 50\,/\mathrm{mm}$, so an alloy whose own boundary density lies below that reference is read as a *weaker* network and its transverse damage is *decelerated*. For 316L ($\lambda_0 = 35\,/\mathrm{mm}$) this gives $f_\lambda(90°) = 0.76$ even though the measured 316L transverse ductility is the lowest of its three orientations—a sign reversal. A natural targeted correction is to normalize the projection to each alloy's own build-direction value,

$$\bar{\lambda}_{\mathrm{rel}}(\psi) = \frac{\lambda_{\mathrm{eff}}(\psi)}{\lambda_{\mathrm{eff}}(0°)} = w_{\mathrm{iso}} + (1 - w_{\mathrm{iso}})\sin^2\psi, \tag{1}$$

so that $\bar{\lambda}_{\mathrm{rel}}(0°) \equiv 1$ and the transverse enhancement becomes the material-independent constant $1/w_{\mathrm{iso}}$. By construction this removes the 316L reversal ($f_\lambda(90°)$ moves from 0.76 to 2.8). We quantify the variant under the *identical* production protocol (single S_0 per material, literature g_s, $k_g = 0.15$, transverse short-ligament f_ξ, the same nine cases and the same leakage-free LOOCV); only the normalization convention of f_λ changes. The global normalization stays the locked production protocol, so every headline statistic of the paper is unchanged; this section reports the two conventions side by side rather than overwriting them.

The correction does exactly what the diagnosis predicts and no more. At the projection-factor level the 316L transverse reversal is removed, and the strongly columnar Ti-6Al-4V is essentially unaffected (calibration 10.76% → 9.41%, blind 10.8% → 9.4%), although its directional ratio now slightly over-corrects (1.65 → 2.06 against the measured 1.67). For the FCC alloys, removing the deceleration merely replaces the transverse under-prediction with an *over-acceleration* of transverse damage: with only a single fitted S_0 per material, the stronger yet now direction-consistent transverse term cannot be absorbed, and both calibration (316L 28.2% → 37.3%) and blind prediction (316L 43.8% → 66.9%) degrade. The nine-case averages confirm the net effect—calibration 20.4% → 22.7% and blind 31.2% → 37.1%—and the information criteria move further against the modified model (ΔAIC +10.8 → +17.9 at $k = 3$). We therefore retain the global normalization as the production protocol and report the relative form as a *diagnostic*, not a competing model.

The value of this side result is that it is a *negative control* on the diagnosis: the worst blind fold was traced to a specific normalization artifact, and correcting that artifact in

Table S9: Public AM dataset hosts assessed for external validation, against the requirement of as-built LPBF tensile data at more than one build orientation for Ti-6Al-4V, 316L, AlSi10Mg and IN718.

Host	**Access**	**Relevant content**	**Verdict for this work**
NIST AM-Bench	public metadata; measured files not retrieved here	macroscale tension for IN625/IN718 only; Ti-6Al-4V challenge is fatigue, AlSi10Mg absent	**no Ti-6Al-4V or AlSi10Mg tensile data**; cannot validate the calibration alloys
NIST repository, HIP Ti-6Al-4V (mds2-3402)	public, NIST Open License; retrieved via the object-store endpoint, sha256 verified	LPBF Ti-6Al-4V, 300 specimens, 2 orientations, 30 published group means, 15 HIP/anneal conditions	**downloaded and integrated** as the 30-point `nist_mds2_3402` source; heat-treatment scope, no 45°
America Makes	members only, US institutions	AM process/property data (CORE)	inaccessible; excluded
Materials Data Facility	public, no account needed	heterogeneous materials data	no compatible AM tensile content
Zenodo / Figshare / Dryad	public (per-item licence)	per-publication deposits, mostly single-orientation	most realistic future source; no consolidated multi-orientation set found
NIMS MatNavi	registration + institutional domain	metallic property database (density, elastic constants, creep)	registration required and mass download/scraping prohibited; not used
Materials Project / OQMD / NOMAD	public or API key	DFT ground-state properties	no tensile or fracture data; not applicable

Table S10: Global (production) versus relative normalization of the melt-pool-boundary projection, identical protocol otherwise. "cal" is the nine-case calibration average and "blind" the leakage-free leave-one-orientation-out average, both in % fracture-strain error; $r_{0/90}$ is the predicted 0°/90° fracture-strain ratio with the measured value in parentheses.

	Error [%]		$r_{0/90}$ (meas.)		
Normalization	cal	blind	Ti-6Al-4V	316L	AlSi10Mg
Global (production)	20.4	31.2	1.65 (1.67)	2.42 (1.45)	2.40 (1.51)
Relative (targeted)	22.7	37.1	2.06 (1.67)	4.30 (1.45)	2.23 (1.51)
Lemaitre baseline	18.6	24.3	$r_{0/90} \equiv 1$ (isotropic)		

isolation removes the qualitative sign reversal yet leaves the aggregate accuracy—and the model-selection outcome—unchanged in direction and worse in margin. It cleanly separates two questions the literature often conflates: whether a descriptor encodes the right *mechanism* (here, after correction, yes at the factor level) and whether it buys out-of-sample *accuracy* (no, at equal parameter count). This reinforces the central claim of the work—the demonstrated contribution is the mapped applicability boundary and the directional information, not a reduction of the blind error. It also motivates a concrete targeted follow-on: a per-alloy *transverse saturation-stress* correction $(1 - \kappa \sin^2 \psi)$ applied to the transverse hardening saturation, designed to let the single S_0 absorb the strengthened transverse term instead of trading one FCC failure mode for another. We tested this follow-on directly (Table S11); it does not rescue the relative form.

Test of the follow-on saturation-stress correction. For each alloy S_0 is fixed on the build direction $\psi = 0°$ alone—where the correction factor is identically unity, so the anchor is independent of κ—and the 90° signed fracture-strain error is then scanned over $\kappa \in [-0.4, +0.4]$ under the same relative normalization and the otherwise-identical production protocol ($k_g = 0.15$, literature g_s, transverse short-ligament f_ξ). Isolating the transverse orientation this way measures precisely the quantity that a single S_0 cannot reach: the *shape* of the transverse term.

The lever acts with the wrong sign for the ductile FCC alloys. Lowering the transverse saturation stress ($\kappa > 0$, the physically suggested direction) accelerates the plastic-flow driving force of damage accumulation ($\dot{p}$) more than it reduces the thermodynamic driving force $Y = \tilde{\sigma}^2/2E$, so $\varepsilon_f(90°)$ falls further and the 316L error deepens from −82.7% to −88.4%. Reversing the sign ($\kappa < 0$) recovers 316L only from −82.7% to −68.6% at the largest admissible $|\kappa| = 0.4$—still about a factor of three short of the measured transverse ductility, and reaching it would demand a transverse saturation-stress increase well outside any physical range. AlSi10Mg is already within ≈ 2% at $\kappa = 0$ and Ti-6Al-4V is already near its measured ratio, so the correction is neutral-to-marginal for both. We therefore conclude that no physically admissible transverse saturation-stress correction lets a single S_0 absorb the strengthened transverse term; the FCC transverse deficiency is intrinsic to the shape of the relative projection (which even mis-anchors the 316L build direction by ≈ 28%), not a normalizable hardening artifact. This is a second negative control and reinforces the retention of the global normalization as the production protocol.

S7. Hierarchical Bayesian view of S_0 and of the applicability gate

To express the material-specificity of the damage strength in a probabilistic form without a computationally prohibitive full MCMC over the explicit-Taylor solver, we fit a Normal–Normal hierarchical model (closed-form empirical Bayes) to the nine leave-one-orientation-out fold values of $\ln S_0$ (three materials × three folds; every fold value is the converged grid+refine optimum of the fold, as produced by the main-text leakage-free LOOCV script): $\ln S_{0,mk} \sim \mathcal{N}(\theta_m, \sigma^2)$, $\theta_m \sim \mathcal{N}(\mu, \tau^2)$.

The estimated within-material (fold) standard deviation is $\hat{\sigma} = 0.22$ in log-units, while the between-material standard deviation is $\hat{\tau} = 2.31$—an order of magnitude larger—so the material-to-material variation of S_0 dominates the fold-to-fold (orientation) variation within a material. This is the Bayesian counterpart of the cross-source non-transferability finding reported in the main-text blind-test sections: S_0 is strongly material-specific, and a point transfer of S_0 across materials or sources is not supported by the data. The posterior predictive distribution for a new material has median $S_0 = 0.589\,\mathrm{MPa}$ and a 95% interval $[0.013, 26.7]\,\mathrm{MPa}$; the median agrees with the cross-material geometric mean used for the main-text Inconel 718 mode-1 extrapolation (0.589 MPa), confirming that the geometric mean is effectively the shrinkage estimate, and the width of the interval explains why any single transferred value produces large errors unless one orientation of the new material is used

Table S11: Transverse saturation-stress correction $(1 - \kappa \sin^2 \psi)$ under the relative normalization. For each alloy S_0 is anchored on $\psi = 0°$ only (the factor is unity there, so the anchor is κ-independent) and the reported $\varepsilon_f(90°)$ signed error [%] isolates the transverse effect of κ. Protocol otherwise as production; a reduced-integrator directional test, to which the conclusion is insensitive because the trend is monotone and the residual stays $\approx 70\%$ at the largest admissible $|\kappa|$.

	meas.	$\varepsilon_f(90°)$ signed error [%]				
Material	$\varepsilon_{f,90}$	$\kappa = -0.4$	-0.2	0	$+0.2$	$+0.4$
Ti-6Al-4V	0.048	$+3.7$	$+1.6$	$+10.2$	$+4.2$	$+4.3$
316L	0.255	-68.6	-76.3	-82.7	-86.4	-88.4
AlSi10Mg	0.037	-2.5	$+3.4$	-0.2	$+6.0$	$+1.6$

to recalibrate. The fold pattern is additionally informative: Ti-6Al-4V yields an identical S_0 in all three folds (stable), whereas 316L varies from 2.0 to 3.2 MPa across folds (orientation-sensitive), showing that the within-material stability itself is material-dependent. The hierarchical structure is reported as a transparent prior, not as a substitute for per-material calibration; the main-text decision procedure quotes its median and 95% interval when the target material has no orientation data available for re-calibration.

Directional-gain MCMC extension (emcee). The closed-form S_0 hierarchy is extended to the *directional* claim with a full affine-invariant MCMC (`emcee`, seed 42; 64 walkers, 12000 steps, 3000 burn-in, thin 15; mean acceptance fraction 0.40). This is a purely post-hoc analysis: it re-uses the stored predictions of the $n = 23$ paired blind set and changes no production parameter. For each blind group with a complete $\{0°, 45°, 90°\}$ triple the group level is removed by within-group centring of $\ln \varepsilon_f$, giving an observed directional contrast o and a model contrast c; a per-group directional gain is then sampled from $o_i = \beta_{g(i)} c_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, s^2)$, with the partial-pooling prior $\beta_g \sim \mathcal{N}(\mu, \tau^2)$, $\mu \sim \mathcal{N}(0, 3^2)$, $\tau \sim \mathrm{HN}(2)$, $s \sim \mathrm{HN}(1)$ (six groups, 18 observations—the complete $\{0°, 45°, 90°\}$ orientation triples available in the $n = 23$ leakage-free blind set; the two remaining groups, a NIST mds2 Ti-6Al-4V 0°/90° pair and the IN718 30°/90° pair, were not pooled because a two-orientation contrast gives a single degree of freedom and a heterogeneous contrast geometry across groups). A gain $\beta_g \approx 1$ means the model reproduces the directional *amplitude* as well as its sign; $\beta_g \approx 0$ means no directional information. Table S12 reports the per-group posterior together with a deterministic directional-capture statistic $R^2 = 1 - \sum_i (o_i - \hat{\beta} c_i)^2 / \sum_i o_i^2$ evaluated on the unshrunk group contrasts (a model with no directional signal gives $R^2 \to 0$, one with the wrong polarity gives $R^2 < 0$).

The pooled gain $\mu = 0.11$ $[-0.38, 0.63]$ brackets zero, so no directional advantage is claimed across all materials jointly—the higher-powered counterpart of the failure-to-distinguish reading on absolute accuracy. Resolving the hierarchy by group recovers the applicability-domain structure: only the columnar Ti-6Al-4V $v12$ group has a positive gain with $R^2_\psi = +0.93$ (the 0°/90° ratio reproduced near-quantitatively, predicted 1.89 against measured 1.67), whereas every FCC or reversed-polarity group has $R^2_\psi < 0$, the same transverse over-projection that the projection-normalization analysis (S6) traces to the $\lambda_{\mathrm{eff}}(90°) = \lambda_0$ saturation. The Lemaitre baseline is not separately gain-fitted: its predicted within-group directional contrast is numerically ≈ 0 (contrast SD 0.003–0.13 against an observed 0.15–0.49), so a slope on a near-zero regressor is not identifiable; its direction-blindness is reported directly through the vanishing contrast and $R^2_\psi \approx 0$ rather than through an unstable β. The head-line 0°/90° ratios for the identifiable columnar group are: measured 1.67, modified 1.89, Lemaitre 0.99 (identically unity up to the solver tolerance). These are held-out blind-set values; the β_g gain is invariant to the single per-material S_0 (which shifts the level, cancelled by the within-group centring), so the gain reflects the orientation shape

Table S12: Per-group directional-gain posterior (modified descriptor model, `emcee`, seed 42). "Gain" is the posterior median of β_g with its 89% highest-posterior-density interval; $P(\beta > 0.5)$ is the posterior probability that the model recovers at least half of the measured directional amplitude; R^2_ψ is the deterministic directional-capture statistic on the group contrasts. A positive, high-R^2 gain appears only in the correctly-polarised columnar Ti-6Al-4V ($v12$) group; the FCC and reversed-polarity groups carry $R^2 < 0$ (transverse over-projection).

Material	**Held source**	**Gain β_g [HPD89]**	$P(\beta>.5)$	R^2_ψ
Ti-6Al-4V	$v12$ original	0.46 [−0.16, 1.18]	0.463	+0.93
Ti-6Al-4V	sun2024adem	0.21 [−0.38, 0.85]	0.222	−0.42
AlSi10Mg	$v12$ original	0.30 [−0.15, 0.76]	0.236	−1.19
AlSi10Mg	awd2018metals	0.11 [−0.37, 0.60]	0.095	−3.38
316L	$v12$ original	0.21 [−0.11, 0.54]	0.076	−8.86
316L	hitzler2017materials	−0.68 [−1.29, 0.07]	0.002	−1.76
Pooled μ (all six groups)		0.11 [−0.38, 0.63] — brackets zero (not robust when pooled)		

carried by the projection alone; the corresponding in-sample production calibration ratio is 1.65 (main-text Table 1). The absolute (level) prediction errors of the two models remain statistically indistinguishable and are reported by the $n = 23$ paired test in the main text, not by this gain analysis.

Operational consequence: the applicability-gated hybrid. The main text (gated-hybrid table) turns the picture above into a per-alloy routing rule keyed on the columnar aspect ratio ξ_0: use the directional descriptor when $\xi_0 \geq 3$ and fall back to the isotropic Lemaitre baseline otherwise, so the model is never worse than the baseline by construction. The gate uses only characterization inputs (ξ_0, or alternatively the melt-pool-boundary density λ_0) and no fracture-strain target, and it does not add a fitted parameter. Table S13 sweeps the gate value and, as an independent metric, the λ_0 alternative, evaluated on the same nine leakage-free LOOCV folds as the main text. The strongly-columnar threshold $\xi_0 \geq 3$ picks exactly the alloy where the directional model wins (Ti-6Al-4V) and reproduces the hindsight per-alloy oracle; loosening it ($\xi_0 \geq 2$) pulls 316L into the directional branch and degrades the hybrid, tightening it ($\xi_0 \geq 4$) removes the descriptor everywhere and reverts to the pure baseline mean. Substituting λ_0 for ξ_0 mis-routes AlSi10Mg (its $\lambda_0 = 60\,\mathrm{mm}^{-1}$ exceeds the reference although its morphology is near-equiaxed) and again pulls the hybrid back toward the pure-model average. With three alloys this is a proof-of-concept for mechanism-driven routing, not a validated operating curve; on new materials both the metric and the threshold should be re-registered against the mechanism check of the main-text decision procedure.

Provenance and safety margin of the $\xi_0 \geq 3$ threshold. The threshold is fixed for a physical reason rather than fitted to the error curve: $\xi_0 \geq 3$ marks where the projection's own premise—a connected, strongly columnar melt-pool-boundary network ($\lambda_{\mathrm{eff}}(90°) = \lambda_0$, S6)—ceases to hold, so the near-equiaxed FCC alloys fall outside the descriptor's physical validity rather than into an unlucky bin. Two properties bound the extrapolation risk. First, the gate is an *asymmetric, fail-safe* rule: the descriptor must be *earned* by positive columnar evidence and any material lacking it defaults to the isotropic baseline, so applying the gate to an unseen alloy degrades at worst to the baseline (the conservative reference used throughout this work), never to a misapplied descriptor. Second, $\theta = 3$ sits on a plateau rather than a knife-edge: over the four materials $\xi_0 \in \{1.5, 2.5, 3.0, 3.5\}$ the routing (Ti-6Al-4V and Inconel 718 to the descriptor, 316L and AlSi10Mg to the baseline), and hence the 15.9% LOOCV and 41.5% external headline errors, is invariant for any $\theta \in (2.5, 3.0]$ (computed in `analyze_gated_hybrid.py`, field `gate_plateau`). The activation branch is corroborated out of calibration by Inconel 718—a

blind material at $\xi_0 = 3.0$ that the gate routes to the descriptor and where the descriptor beats the baseline externally (57.2 versus 86.0, Table S14). These bound but do not remove the risk: with four materials the plateau is set by a single ξ_0 gap and the operating curve is not statistically validated, so on a new alloy the metric and threshold should be re-registered against the mechanism check of the main-text decision procedure, defaulting to the baseline whenever ξ_0 lies near the $(2.5, 3.0]$ boundary.

Operational recipe for applying the gate to a new alloy. The gate is intended to be executed from characterization inputs alone and to default to the safe branch under doubt. *(1) Measure* the columnar grain aspect ratio ξ_0—the mean grain length-to-width ratio along the build direction from EBSD or optical micrographs (equiaxed grains give $\xi_0 \approx 1$)—and, as a secondary check, the melt-pool-boundary line density λ_0. *(2) Test the activation branch*: the directional descriptor is used only if *both* ξ_0 is clearly at or above the plateau ($\xi_0 \geq 3$, with margin from the 2.5 lower edge) *and* the measured direction trend is consistent with the columnar-acceleration mechanism (main-text decision step 2). *(3) Fail-safe default*: if ξ_0 falls in or near the $(2.5, 3.0]$ boundary band, if the morphology is mixed, or if the mechanism check is ambiguous, the alloy is routed to the isotropic baseline and reported as screening-only—the prescribed behaviour, not a failure, because the baseline is the conservative reference. *(4) Re-register the boundary, not the constant*: $\theta = 3$ is a domain marker for the present four-material span, not a transferable law; when a new family with at least one directional datum becomes available, the boundary should be re-derived by the same leakage-free LOOCV attribution used here—locate the gap between the descriptor-losing and descriptor-winning populations of ξ_0 (in this work the gap between 316L at 2.5 and Inconel 718 at 3.0) and place the threshold inside it. Because ξ_0 enters no fracture fit, the recipe adds no fitted parameter and preserves the leakage-free property of the main-text protocol; the routing uncertainty from measurement scatter in ξ_0 is absorbed by the step-3 fail-safe default.

External validation of the gate on the frozen $n = 23$ blind set. The routing rule is re-applied, unchanged and leakage-free (it uses only the characterization ξ_0), to the independent frozen external blind set, by selecting per point the directional or the isotropic prediction according to the same $\xi_0 \geq 3$ gate (Table S14). The gated hybrid then attains a mean absolute fracture-strain error of 41.5%, below both the always-isotropic baseline (48.3%) and the ungated descriptor (53.1%). The descriptor is *ahead* on both gated-in columnar materials (Ti-6Al-4V 26.8 versus 35.4 on 8 points, Inconel 718 57.2 versus 86.0 on 3 points) and the ungated aggregate only reverses because of the six out-of-domain reversed-polarity 316L points (106.2 versus 58.6) that the gate routes to the baseline; restricted to the gate's own domain ($\xi_0 \geq 3$, $n = 11$) the descriptor leads by 14 points (35.1 versus 49.2). As in the LOOCV attribution, the gate's margin over the baseline is supplied entirely by the descriptor's in-domain accuracy: for the out-of-domain alloys the gate reproduces the baseline identically and adds nothing of its own. These are per-point re-tabulations of the stored frozen blind predictions (no refit, no new solver run); as with the LOOCV gate, the domain is fixed on three alloy families plus Inconel 718 and the threshold is not validated over a wider microstructural span.

S8. Superseded parametrization with orientation multipliers

For traceability we record the earlier parametrization that preceded the leakage-free production protocol reported in the main text. The production model carries *no* empirical orientation multiplier: all direction dependence is generated by the closed-form projection laws and the mean Schmid factor (main-text Section 2). The present section documents the superseded alternative for completeness only.

In that earlier parametrization each material carried three fitted parameters—the damage energy strength S_0 plus two residual orientation multipliers a_{45} and a_{90}, applied as piecewise factors on the damage driving force at the oblique ($\psi = 45°$) and transverse ($\psi = 90°$) orientations—calibrated on the fracture-strain data of all three orientations. Because three pa-

Table S13: Sensitivity of the gated-hybrid mean LOOCV blind fracture-strain error [%] to the gate value and the gate metric. Baseline (all isotropic) is 24.34% and the pure directional model is 31.22%; the hindsight per-alloy oracle is 15.89%. Numbers from `analyze_gated_hybrid.py` / `experiment_gated_hybrid_results.json`.

Gate	**Routed to directional**	**Hybrid mean blind [%]**
$\xi_0 \geq 4$	none	24.34 (reverts to baseline)
$\xi_0 \geq 3$	Ti-6Al-4V	**15.9** (matches oracle)
$\xi_0 \geq 2$	Ti-6Al-4V, 316L	23.20
$\lambda_0 \geq \lambda_{\text{ref}}$	Ti-6Al-4V, AlSi10Mg	23.92
Hindsight oracle	per-alloy winner	15.89

Table S14: External validation of the applicability gate on the frozen $n = 23$ blind set (mean absolute fracture-strain error [%] per material; gate branch from the characterization ξ_0 only). Gated-hybrid mean 41.5% < always-isotropic 48.3% < ungated descriptor 53.1%. Numbers from `analyze_gated_hybrid.py` / `experiment_gated_hybrid_results.json` (external block).

Material	ξ_0	**Gate branch**	Descriptor	Baseline
Ti-6Al-4V	3.5	descriptor	**26.8**	35.4
Inconel 718	3.0	descriptor	**57.2**	86.0
316L	2.5	baseline	106.2	**58.6**
AlSi10Mg	1.5	baseline	32.9	**36.3**
Ungated mean (23 pts)			53.1	48.3
In-domain ($\xi_0 \geq 3$, 11 pts)			**35.1**	49.2
Gated hybrid (23 pts)			**41.5**	

rameters are fitted to the three measured orientations of each material, the resulting average calibration fracture-strain error of 0.76% over the nine cases is a near-interpolation result and is *not* a predictive error; it is quoted here only to explain why the main-text single-parameter calibration error (20.35%) is larger yet honest. A fixed-global-parameter sensitivity check (orientation multipliers and g_s fixed at values calibrated on all nine cases) yielded an average leave-one-orientation-out error of 2.44%, but that figure is likewise not a blind prediction because the held-out orientation indirectly informed the fixed parameters; it is therefore re-labelled a *fixed-global-parameter sensitivity check* and is not reported as validation anywhere in this work. The calibrated multiplier values (Ti-6Al-4V: $a_{45} = 1.246$, $a_{90} = 2.740$; 316L SS: 1.048, 1.231; AlSi10Mg: 0.558, 0.563) are recorded here for reproducibility. The leakage-free protocol of the main text removes these multipliers entirely (fixed at unity, hence not model parameters) and is the only protocol on which the reported calibration and blind cross-validation results are based.

S9. Open-source CPFEM framework availability in the present environment

The coupled-damage full-field references of the main text are produced with in-house solvers, which raises the question of whether an established open-source framework could be used instead. The answer was established by measurement rather than by assumption, on the machine on which all results of this work were produced (Windows 11 (10.0.26200), CPython 3.13.0, pip 26.1.2, conda 26.1.1; pip configured to a PyPI mirror). All probes are read-only: `pip index versions`, `pip install -dry-run` (which resolves and inspects distributions without writing into `site-packages`) and `conda search -platform win-64`. Twelve distribution names were probed; the outcome is summarized in Table S15.

Two distinctions in the table are deliberate, because they separate "this cannot be done here" from "this was not done here", and only the second would be a fair criticism of the paper. (i) MOOSE and PRISMS-PF are blocked

Table S15: Availability of open-source CPFEM / phase-field frameworks in the environment in which all results of this work were produced, from read-only pip and conda probes.

Framework	**Probed names**	**PyPI**	**conda-forge** `win-64`
INL MOOSE (FE multiphysics)	`pymoose`, `moose`	`pymoose` resolves but is the *neuroscience* MOOSE; `moose` is an unrelated package with invalid metadata	not found
FEniCSx solver	`fenics-dolfinx`	no matching distribution	0.9.0 exists, Python 3.9–3.12 only (no 3.13)
FEniCS legacy solver	`dolfin`	no matching distribution	–
FEniCS legacy meta-package	`fenics`	resolves to the form compilers only (FFC/FIAT/UFL/Dijitso), no assembly or solver layer	–
PRISMS-PF	`prisms`, `prisms-pf`, `pyprisms`	no matching distribution	not found (only the unrelated `prisms-jobs`)
DAMASK	`damask`, `damask-core`, `damask-parse`	`damask` 3.1.0 already installed	`python-damask` up to Python 3.10

by the *platform*: no Windows distribution exists at any Python version. The two PyPI names that carry the name MOOSE are not the Idaho National Laboratory finite-element framework at all—one is a computational-neuroscience simulator, the other an unrelated package whose packaging metadata is invalid—and conda-forge publishes no `win-64 moose`; likewise `prisms`, `prisms-pf` and `pyprisms` are absent from both indices for Windows. (ii) FEniCS is blocked only by the *interpreter version*: `fenics-dolfinx` 0.9.0 has `win-64` conda-forge builds for Python 3.9–3.12 but none for 3.13, and the legacy `dolfin` solver has no PyPI distribution at all (the `fenics` meta-package resolves only to the form compilers, without an assembly or solver layer). Using FEniCS would therefore require maintaining a second conda environment pinned to Python ≤ 3.12 alongside the 3.13 stack on which every other result depends. DAMASK 3.1.0 is installed, but it supplies the spectral-FFT solver `DAMASK_grid` and mesh utilities only—no finite-element solver—and its built-in damage model diverges at damage activation in this setting (the main-text full-field section); it therefore enters this work exclusively as an external *damage-free* full-field reference and never as a host for the damage model.

A framework migration would also not remove the verification burden. MOOSE, FEniCS and PRISMS-PF supply mesh handling, assembly, linear algebra, parallelism and time integration, but none of them supplies the constitutive content of this work: the Taylor-type aggregate homogenization with per-grain crystal plasticity, the AM-corrected Lemaitre driving force with its five descriptors and projection laws, the plastic-strain activation gate, and the aggregate fracture criterion. Adopting a framework would change the discretisation, not the physics, and every constitutive line—and with it the entire verification programme (unit tests, convergence studies, analytic limits) reported in the main-text code-verification section—would still have to be written and validated. That is the sense in which the framework question and the code-verification question are independent, and it is why the verification reported here is carried out on the in-house solver.

One external finite-element engine is nonetheless installable here by `pip`: `scikit-fem`, a pure-Python library that supplies mesh handling, shape functions, Gaussian quadrature, sparse assembly and linear solvers but no crystal-plasticity or damage constitutive content. It was used to re-drive the identical coupled-damage law as an independent cross-check of the two-dimensional in-house reference; the agreement on Ti-6Al-4V and the honest disagreement on AlSi10Mg are reported in Section S13. This does not change the conclusions of the present section:

platform-level fully coupled solvers (MOOSE, FEniCSx, PRISMS-PF) remain uninstallable in the Windows production environment, and a phase-field calculation is still future work. FEniCSx was nonetheless brought up afterwards in a separate Linux `conda-forge` environment to supply a stronger independent re-implementation cross-check, reported in Section S13; this reinforces rather than alters the conclusions above.

S10. Parameter provenance and degrees of freedom

Table S16 gives the class-by-class provenance and fitted degrees of freedom of the full parameter set quoted in main-text Table 1. Each parameter belongs to exactly one class, and the only free parameters fitted to fracture data are the three damage energy strengths S_0 (one per material); both the descriptor-based model and the equal-parameter Lemaitre baseline therefore carry the *identical* fitted count, which is what makes the leave-one-orientation-out comparison leakage-free.

S11. Expanded leakage-free paired blind set and statistical upgrade

This section reports the full per-point result behind the main-text expanded paired blind table (`tab:blind_expanded`) and the statistical treatments built on it, generated by the archived drivers `blind_eval_expanded.py`, `stats_upgrade_expanded.py`, `statistical_upgrade_bayes.py` and their result JSONs. The expanded paired blind set lifts the headline descriptor-versus-baseline comparison from the nine forced LOOCV folds to $n = 23$ independent held-out *as-built* points. Every point is predicted blind under the identical single-S_0 leakage-free protocol; the Lemaitre baseline ($f_{\mathrm{AM}} \equiv 1$) is refit fold-by-fold on the same training set, so both models share the parameterization, fitting freedom and data. The baseline S_0 search uses the reduced grid $[0.05, 0.2, 0.8, 3.2]$ with local refinement: because the fracture strain is monotone-increasing in S_0 under $f_{\mathrm{AM}} \equiv 1$, the two omitted top grid points (12.8, 50.0) are provably non-optimal and the selected S_0 is identical to the full-grid argmin. The 30 NIST mds2-3402 HIP heat-treated rows are excluded from the headline count as an out-of-domain diagnosis.

Mixed-effects treatment. The 23 points were stacked long (one row per model) and a random-intercept model $|e| \sim$ `model` $+ (1 \mid \mathrm{group})$ was fitted with a material- and a source-clustering. With material as the grouping factor the descriptor-minus-Lemaitre fixed effect is +4.81 points (SE 15.74, $p = 0.760$, 4 groups, random-intercept SD 21.25 points); with source as the grouping factor it is +4.81 points (SE 14.99, $p = 0.748$, 6 groups, random-intercept SD 0.0 points). A Kruskal–Wallis test on the per-material paired differences (316L +47.7, AlSi10Mg −3.3, IN718 −28.8, Ti-6Al-4V −8.6 points) gives $p = 0.367$: no material shows a reliably different descriptor-versus-baseline gap, consistent with the overall non-separation. The point of the mixed-effects fit is precisely that the descriptor effect is null once the clustering is accounted for, while the raw per-point spread is source-dominated.

Hierarchical shrinkage of the paired gap. Extending the closed-form Normal–Normal empirical-Bayes hierarchy of S7 to the 23-point paired gap, the grand-mean difference is +1.72 points (within-material SD 59.2, between-material SD 44.5; positive = descriptor worse); the shrunken per-material means (316L +37.2, AlSi10Mg −2.2, IN718 −17.5, Ti-6Al-4V −6.7) all move toward zero. The posterior predictive for a *new* material is 90% interval $[-120.1, +123.6]$ points, which brackets zero: a new alloy is expected to show no reliable descriptor advantage in blind accuracy, the probabilistic counterpart of the failure-to-distinguish reading. A full MCMC over the explicit-Taylor solver remains computationally infeasible, so the closed-form Normal–Normal treatment is retained and stated as such rather than approximated by an unsupported sampler.

Orientation-ranking (directional) metric. For each (material, held-source) group with ≥ 2 orientations the Kendall τ between measured

Table S16: Parameter provenance and degrees of freedom under the leakage-free production protocol. "Measured" descriptors are literature-reported microstructural measurements; "literature-fixed" constants follow published elastic/Voce data; "calibrated" parameters are the only free parameters fitted to fracture data (one S_0 per material); "assumed/prescribed" values follow standard practice or stereological assumptions and are never fitted; both models carry the *identical* fitted count (3).

Class	**Parameters**	**Source**	**Fitted DOF**
Measured (literature)	ϕ, D_0, λ, ξ, θ, and D_c	XCT/EBSD/OM literature values (main-text experimental-sources section); D_c fixed by the measured fracture load drop, not a least-squares fit	0
Literature-fixed	C_{ij}, g_0, g_s, h_0, a, n, $\dot{\gamma}_0$, q	published elastic/Voce data	0
Calibrated	S_0 (one per material)	least-squares on ε_f (main-text calibration section)	1 per material (3 total)
Assumed/ prescribed	$s = 1$, p_D, β_{1-6}, $m_{1,2}$, $k_g = 0.15$, $w_{\text{iso}} = 0.25$, ϕ_{crit}	standard exponents / stereological assumptions (p_D a prescribed damage-initiation threshold, never fitted)	0
Total			**3 (both models)**

Table S17: Expanded leakage-free paired blind set ($n = 23$): per-point measured fracture strain $\varepsilon_f^{\text{exp}}$, modified and Lemaitre blind predictions, and absolute errors in %. Material counts: Ti-6Al-4V 8, 316L 6, AlSi10Mg 6, IN718 3. Modified mean/median 53.1%/34.7% (5/23 within 10%); Lemaitre 48.3%/45.0% (2/23).

			$\varepsilon_f^{\text{exp}}$	**Modified**		**Lemaitre**	
Material	**Source**	ψ [°]		pred.	err /%	pred.	err /%
316L	hitzler2017materials	0	0.1176	0.5419	360.8	0.2659	126.1
316L	hitzler2017materials	45	0.3256	0.3902	19.9	0.2554	21.6
316L	hitzler2017materials	90	0.3324	0.2236	32.7	0.2047	38.4
316L	v12 original	0	0.3700	0.1427	61.4	0.1672	54.8
316L	v12 original	45	0.3100	0.0782	74.8	0.1303	58.0
316L	v12 original	90	0.2550	0.0310	87.9	0.1206	52.7
AlSi10Mg	awd2018metals	0	0.0830	0.0688	17.2	0.0457	45.0
AlSi10Mg	awd2018metals	45	0.0570	0.0373	34.7	0.0459	19.5
AlSi10Mg	awd2018metals	90	0.0810	0.0286	64.7	0.0460	43.2
AlSi10Mg	v12 original	0	0.0560	0.0975	74.0	0.0615	9.8
AlSi10Mg	v12 original	45	0.0470	0.0498	6.0	0.0620	31.8
AlSi10Mg	v12 original	90	0.0370	0.0366	1.1	0.0623	68.3
IN718	watring2020addma	30	0.2800	0.1312	53.2	0.0751	73.2
IN718	watring2020addma	90	0.0600	0.0406	32.3	0.1398	133.1
IN718	watring2020addma	90	0.2900	0.0406	86.0	0.1398	51.8
Ti-6Al-4V	nist mds2 3402	0	0.1680	0.0723	57.0	0.0577	65.7
Ti-6Al-4V	nist mds2 3402	90	0.1928	0.0439	77.2	0.0581	69.9
Ti-6Al-4V	sun2024adem	0	0.0934	0.0924	1.1	0.0577	38.2
Ti-6Al-4V	sun2024adem	45	0.0533	0.0646	21.3	0.0580	8.9
Ti-6Al-4V	sun2024adem	90	0.0789	0.0490	37.9	0.0581	26.4
Ti-6Al-4V	v12 original	0	0.0800	0.0924	15.4	0.0709	11.4
Ti-6Al-4V	v12 original	45	0.0630	0.0646	2.6	0.0715	13.5
Ti-6Al-4V	v12 original	90	0.0480	0.0490	2.1	0.0717	49.4

and predicted fracture-strain ordering was computed for both models.

Table S18: Per-group Kendall orientation-ordering correlation (positive = measured ductility order recovered). Descriptor mean $\bar{\tau} = +0.208$ (median $+0.333$), Spearman mean $+0.25$, monotone $0° > 45° > 90°$ hits 6/6; Lemaitre mean $\bar{\tau} = -0.208$ (median -0.333), i. e. a tendency to reverse the order.

Material	Held source	n_ψ	$\bar{\tau}_{\mathrm{desc}}$	τ_{lem}
316L	hitzler2017materials	3	-1.00	-1.00
316L	v12 original	3	+1.00	+1.00
AlSi10Mg	awd2018metals	3	+0.33	-0.33
AlSi10Mg	v12 original	3	+1.00	-1.00
IN718	watring/hovig	2	+0.00	+0.00
Ti-6Al-4V	nist mds2 3402	2	-1.00	+1.00
Ti-6Al-4V	sun2024adem	3	+0.33	-0.33
Ti-6Al-4V	v12 original	3	+1.00	-1.00

The descriptor model carries a positive mean orientation correlation whereas the isotropic baseline is symmetrically negative; this is the direction information that the mean-accuracy comparison cannot resolve and that the main text identifies as the descriptor model's only material advantage. To quantify its statistical resolution we bootstrap the eight source–material groups (20,000 group resamples, seed 42): the descriptor mean is $\bar{\tau} = +0.208$ with 95% CI $[-0.333, +0.708]$ and a sign-flip permutation $p = 0.53$, the Lemaitre mean $\bar{\tau} = -0.208$ with CI $[-0.750, +0.333]$, $p = 0.51$, and the paired difference $\bar{\tau}_{\mathrm{desc}} - \bar{\tau}_{\mathrm{lem}} = +0.417$ with CI $[-0.417, +1.250]$, $p = 0.43$. All three intervals bracket zero, so the directional advantage is *positive but not individually significant* at eight groups—reported as an honest point estimate whose near-quantitative recovery is localized to the correctly-polarised columnar sub-domain rather than claimed across the full set.

Information criteria on the nine blind folds (auxiliary point estimate). The main text scores both models on the *blind* $n = 9$ leave-one-orientation-out folds under the identical 20-grain Taylor aggregate, identical per-fold training data and identical fitted-parameter count $k = 3$ (one S_0 per material); the AM descriptor coefficients and the Voce saturation stresses are fixed by characterization/literature and are not counted as fitted degrees of freedom for either model. This $k = 3$ convention is the single counting rule used consistently for every information-criterion value in the paper, and a deliberately conservative $k = 7$ alternative that also counts the assumed coupling coefficients is quoted only as a bracket and never mixed with the $k = 3$ figures. Under the Gaussian-error approximation with residuals $r_i = |\varepsilon_{f,i}^{\mathrm{exp}} - \varepsilon_{f,i}^{\mathrm{pred}}|$ and variance $\sigma^2 = \frac{1}{n}\sum_i r_i^2$ the log-likelihood is

$$\ln \mathcal{L} = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n} r_i^2, \qquad (2)$$

with $\mathrm{AIC} = 2k - 2\ln\mathcal{L}$ and $\mathrm{BIC} = k\ln n - 2\ln\mathcal{L}$. With $k = 3$ for both models this gives AIC $= -13.5$ and BIC $= -12.9$ for the modified model versus AIC $= -24.3$ and BIC $= -23.7$ for the baseline ($\Delta\mathrm{AIC} = \Delta\mathrm{BIC} = +10.8$; RMS 8.21×10^{-2} versus 4.50×10^{-2}), so the information criteria favour the *Lemaitre baseline* at equal parameter count and equal data: the modified model's blind residuals are dominated by the over-projected 90° folds of 316L (58.9%) and AlSi10Mg (22.7%), and the corrected projection—while physically required—degrades the out-of-sample likelihood sufficiently to reverse the model-selection outcome. This reversal is the honest consequence of removing the sign-inverted projection that had artificially inflated the modified model's blind performance. These criteria remain an auxiliary descriptive point estimate and are not treated as decisive model-selection evidence; the paired, bootstrap and effect-size analyses of this section are the primary evidence.

Effect size and power. With $n = 23$ the paired SD is 61.4 points, Cohen's $d_z = +0.08$, and the minimum detectable mean difference at $\alpha = 0.05$ two-sided and 80% power is ≈ 35.8 points—an order of magnitude above the observed +4.8-point difference, which is nominally in favor of the *baseline* (the descriptor is, if taken at face value, slightly worse rather than better). The paired frequentist tests are non-committal: Wilcoxon signed-rank $p = 1.00$ and a sign test (eleven positive, twelve negative) $p = 1.00$, and a 100 000-sample boot-

strap of the paired mean difference yields a 95% CI of [−16.9, +32.1] points that includes zero. Because a non-rejection of the null is not itself evidence of equivalence, the comparison was examined with two further routes (driver `statistical_upgrade_bayes.py`). A Jeffreys–Zellner–Sioukas Bayes factor on the paired difference (mean +4.81, SD 61.4, SE 12.8 points, $n = 23$), with the standardized-effect index given a Cauchy prior of width 0.2/0.35/0.5/1.0, yields $\mathrm{BF}_{01} = 1.80/2.49/3.23/5.84$—only *weak-to-moderate* evidence *for* no mean-accuracy difference (crossing the Kass–Raftery $\mathrm{BF}_{01} > 3$ grade solely at the two widest priors, and < 2 at the narrowest); a two-one-sided equivalence test (TOST) on the same paired differences *fails to reject* non-equivalence at every margin examined—$p_{\mathrm{TOST}} = 0.345/0.217/0.124/0.064$ for the $\pm 10/\pm 15/\pm 20/\pm 25$-point margins—so equivalence is *not* affirmatively established even at the widest margin. Taken together, the correct reading is a *failure to distinguish* the two models at the resolution the $n = 23$ experiment affords—not a demonstration of equivalence. The information criteria of the main text remain an auxiliary descriptive point estimate and are not treated as decisive model-selection evidence.

Required sample size (a data limit, not a protocol defect). Inverting the same two-sided paired-power calculation at the observed paired SD $\sigma_d = 61.4$ points, the number of paired blind points required to resolve a mean descriptor-versus-baseline difference Δ at $\alpha = 0.05$ and 80% power is $n = \left((z_{0.975} + z_{0.8})\,\sigma_d/\Delta\right)^2$ with $z_{0.975} + z_{0.8} = 2.80$ (Table S19). The observed +4.8-point difference would need ≈ 1285 points; even a 10- or 20-point engineering margin needs ≈ 296 or ≈ 74 points. As a self-consistency check the formula returns $n \approx 23$ for $\Delta = 35.8$ points, recovering the minimum detectable effect at the actual sample size. Against this requirement only six complete leakage-free as-built $\{0°, 45°, 90°\}$ triples (≈ 23 points) exist, so the power shortfall is quantified and is a field-wide data limit rather than a defect of the protocol.

Blind-set ceiling (data-limited, not method-limited). The expanded set is capped by the public data rather than by the protocol: among the orientation-resolved *as-built* sources only six (source, material) groups supply a complete leakage-free $\{0°, 45°, 90°\}$ triple (AWD AlSi10Mg, Hitzler 316L, Sun Ti-6Al-4V and the three `v12`-held folds), which is also the group count entering the directional-gain pooling of S7. The remaining candidates are heat-treated and therefore out-of-domain under the same as-built constraint—the 30 NIST mds2-3402 rows are HIP and the Watring IN718 set is only $\{30°, 90°\}$. Raising n further would require new experiments or relaxing the leakage-free as-built rule; neither is done here, so every Bayes-factor, TOST and Kendall conclusion above inherits this six-triple ceiling.

Table S19: Paired-sample-size requirement to resolve a mean descriptor-versus-baseline difference Δ at $\alpha = 0.05$ two-sided and 80% power, using the observed paired SD $\sigma_d = 61.4$ points: $n = \lceil((z_{0.975} + z_{0.8})\,\sigma_d/\Delta)^2\rceil$, $z_{0.975} + z_{0.8} = 2.80$.

Δ [points]	required paired n
4.8 (observed)	1285
10	296
15	132
20	74
35.8 (MDE, self-check)	≈ 23

S12. Global sensitivity, β_4 robustness and model-simplification invariance

This section reports the variance-based sensitivity decomposition and the two robustness checks behind the main-text parameter-sensitivity paragraph, generated by `sensitivity_sobol_beta4.py` and `model_simplification.py` and their result JSONs.

Sobol-type decomposition from the 3^3 factorial. Re-casting the 27-run full-factorial design over β_3, β_4, β_5 as a balanced ANOVA and splitting the total sum of squares (SST = 0.02237) into main and two-way interaction terms gives the first-order and total-effect indices in Table S20. The response variance

is carried almost entirely by β_4 ($S_1 = 0.989$, $T = 0.991$); β_5 is minor ($S_1 = 0.008$), β_3 negligible ($S_1 \approx 0.000$), and the directional coefficient k_g contributes a zero marginal span at 0°.

Table S20: Variance-based (Sobol-type) indices from the balanced ANOVA of the 3^3 full-factorial grid (Ti-6Al-4V, 0°). First-order S_1 and total-effect T indices; k_g reported separately as its marginal span.

Parameter	Main SS /%	S_1	T
β_3	0.03	0.000	0.000
β_4	98.87	0.989	0.991
β_5	0.84	0.008	0.011
k_g	marginal span 0.0% at 0°		

Paired-blind robustness to β_4 over its ±40% range. Because β_4 dominates the absolute response, the question is whether its assumed value could overturn the leakage-free conclusion. Table S21 re-runs the $n = 23$ paired blind comparison at 0.6×, 1.0× and 1.4× each material's production β_4. The paired mean difference shifts with β_4 (from −1.5 to +17.2 points), but the bootstrap CI *contains zero at every level* and no p-value approaches significance: β_4 dominance translates the absolute descriptor error without creating a resolvable modified-versus-baseline separation.

Model-simplification invariance. Guided by the Sobol ranking, the three inert sub-terms (β_3, β_5, k_g) were individually and jointly deactivated and the headline $n = 23$ blind metrics re-measured (Table S22). Deactivation moves the mean blind error by at most 2.6 points and the within-10% count by at most one—far below the 35.8-point MDE—so these terms can be fixed rather than carried as tunable hidden degrees of freedom. After fixing them the effective production parameter set is the single fitted S_0 per material plus an assumed block in which only β_1, β_2 and β_4 are active, an explicit reduction of the hidden-degree-of-freedom surface reported in S10.

Factorial interaction, k_g and D_c detail. In the 3^3 full-factorial design the β_4 marginal means span 0.0587 → 0.1251 (91.9% of the baseline), β_5 0.0822 → 0.0887 (8.9%) and β_3 1.6%; two-factor interactions reach 92.9% ($\beta_3 \times \beta_4$, itself dominated by the β_4 main effect), 15.8% ($\beta_4 \times \beta_5$) and 9.0% ($\beta_3 \times \beta_5$). The directional-strength coefficient k_g does not enter at 0° ($\hat{g}_s(0°) = g_s$, 0.0% span) and at 90° spans 0.4% over $k_g \in [0, 0.4]$ (fracture strain 0.0439 → 0.0441), confirming it as a low-sensitivity assumed parameter. The critical damage threshold D_c over $[0.25, 0.59]$ (±40% of the Ti-6Al-4V value 0.42) has a 0.0% span (0.07229 at every level): fracture is set by the damage avalanche—the failed-grain mass fraction D_{agg} jumps from below to above any D_c in this range within a single strain increment—so D_c is a non-influential threshold parameter in the explored band.

Robustness of the two remaining ξ_{eff}-coupled assumed constants (w_{iso}, β_5). Beyond β_4, the corrected grain-morphology projection enters the response through two further *assumed* (never fitted) constants: the pore-free normalisation w_{iso}, which sets the transverse λ_{eff} contrast, and the short-ligament coefficient β_5, which scales $f_\xi = 1 + \beta_5(1/\xi_{\text{eff}} - 1)$. Both are ξ_{eff}-coupled, so the corrected projection changes their influence and their robustness was re-swept with the single fitted S_0 held at its production value; the reported statistic is the in-sample three-orientation calibration average (a calibration proxy, not a blind prediction). Tables S23 and S24 give the sweep for the two FCC alloys, whose transverse over-projection the main text diagnoses; the strongly columnar Ti-6Al-4V is at its verified optimum ($\beta_5 = 0.2$) and is not re-swept.

The sweep confirms the direction of the main-text diagnosis and, read honestly, qualifies the one-fitted-parameter claim. For 316L the calibration error is sensitive to both constants exactly as the transverse over-projection predicts: raising w_{iso} from 0.25 to 0.70 (weakening the assumed transverse λ_{eff} contrast) drops the 316L calibration error from 28.2% to 8.6%, and lowering β_5 from its production 0.15 to 0.10 drops it from 28.2% to 22.6%, with the 90° calibration component alone falling to 2.3%. AlSi10Mg is much less sensitive: w_{iso} has a shallow minimum near 0.35 (16.6%) and wors-

Table S21: β_4 sweep on the $n = 23$ paired blind set (per-material production value scaled by the factor). Bootstrap 95% CI of the paired mean difference includes zero at every level.

β_4 **factor**	**diff /pt**	**Wilcoxon** p	**bootstrap 95% CI**	d_z
0.6×	−1.47	0.800	[−19.6, +15.5]	−0.034
1.0×	+4.81	1.000	[−17.0, +31.9]	+0.078
1.4×	+17.21	0.410	[−10.1, +50.9]	+0.225

Table S22: Model-simplification invariance on the $n = 23$ paired blind set. Δ values are relative to the full production model (mean 53.1%, median 34.7%, 5 within 10%).

Variant	mean /% (Δ)	median /% (Δ)	pass@10 (Δ)
production	53.1 (–)	34.7 (–)	5 (–)
deactivate β_3	53.7 (+0.6)	33.7 (−0.9)	5 (0)
deactivate β_5	50.7 (−2.4)	29.1 (−5.6)	5 (0)
deactivate k_g	51.4 (−1.7)	34.9 (+0.2)	5 (0)
deactivate $\beta_3 + \beta_5$	51.5 (−1.6)	28.3 (−6.4)	6 (+1)
reduced (all inert)	50.5 (−2.6)	34.0 (−0.7)	6 (+1)

Table S23: Calibration-average error (% fracture strain) against the assumed pore-free normalisation w_{iso}, with S_0 fixed at the production value (316L 3.2 MPa, AlSi10Mg 0.2 MPa). The production value $w_{\mathrm{iso}} = 0.25$ is marked.

	w_{iso}				
Material	0.25 (**prod**)	0.35	0.45	0.55	0.70
316L	**28.2**	20.1	17.5	12.4	8.6
AlSi10Mg	**22.1**	16.6	22.5	26.5	30.8

Table S24: Calibration-average error (% fracture strain) against the assumed short-ligament coefficient β_5, S_0 fixed at the production value. The production β_5 (316L 0.15, AlSi10Mg 0.10) is marked. For 316L the 45° component is invariant at 25.89% across the whole sweep because the corrected projection gives $\xi_{\mathrm{eff}}(45°) = 1$ and hence $f_\xi(45°) = 1$ independently of β_5; the β_5 sensitivity is carried entirely by 0° and 90°.

Material	$\beta_5 \mapsto$ calibration-average error [%]
316L	0.02 → 24.3, 0.04 → 22.5, 0.06 → 24.0, 0.08 → 24.2, 0.10 → 22.6, 0.12 → 27.9, **0.15 (prod)** → **28.2**
AlSi10Mg	0.01 → 20.6, 0.02 → 20.8, 0.03 → 21.0, 0.04 → 21.1, 0.05 → 21.3, 0.07 → 21.6, **0.10 (prod)** → **22.1**

ens beyond it, and β_5 is nearly flat over its whole sweep (20.6–22.1%). The residual FCC transverse error is therefore substantially set by two *assumed* descriptor constants rather than by the single fitted S_0—a qualification we state rather than conceal. We deliberately do *not* re-tune them: they are held at their physically-motivated production values, these are in-sample calibration proxies and not blind predictions, and the main-text applicability statement (the FCC transverse limitation) is unchanged. What the sweep adds is a quantified attribution—the residual maps onto specific assumed constants and hence onto follow-on stereological measurement, not onto a hidden fitted degree of freedom.

S13. External finite-element engine cross-checks (`scikit-fem` and FEniCSx)

This section reports the third-party cross-check referenced in the main-text code-verification and full-field sections. Because MOOSE and PRISMS-PF are blocked at the platform level and `fenics-dolfinx` has no `win-64` build at the interpreter version of the production environment (Section S9), and DAMASK supplies no finite-element solver, the only externally maintained finite-element engine that could be installed there by `pip` is `scikit-fem` 12.0.2,

a pure-`pip` library providing mesh handling, shape functions, Gaussian quadrature, sparse assembly and linear solvers but no crystal-plasticity or damage constitutive content. The identical coupled-damage law—the batch explicit integration copied verbatim from the in-house two-dimensional CPFEM, with the same fixed random seed (42), 16-grain orientation set, elastic stiffness, Voce hardening, AM-corrected Lemaitre driving force, plastic-strain activation gate and fracture criteria—was therefore driven by the external engine on the plane-strain 4×4 four-node (Q4) RVE (16 elements, 64 integration points), with a Newton iteration and line search on a constant tangent elastic preconditioner at $\Delta\varepsilon = 2 \times 10^{-3}$, $n_{\mathrm{sub}} = 30$. A fully independent re-implementation on FEniCSx—installed subsequently in a separate Linux `conda-forge` environment, and probing the re-derivation of the constitutive physics as well as the discretisation and linear solve—is reported at the end of this section.

The three Ti-6Al-4V build orientations reproduce the in-house localisation-onset fracture strains to within 11–17% and preserve the $0° > 45° > 90°$ monotone trend and the same failure criterion, so the Taylor-homogenization localisation signature reported in the main text is not an artefact of the in-house discretisation: an independently maintained finite-element assembly and linear solver returns the same qualitative result and a quantitatively close one. The AlSi10Mg case is reported as a genuine disagreement rather than suppressed: in the more uniform AlSi10Mg the two engines activate the aggregate D_c threshold at different times, a known mesh- and solver-sensitivity of a localisation test on a coarse 16-element RVE, consistent with the main-text statement that the robust quantities are the *sign* and the *orientation trend* rather than a single converged magnitude. This is a framework-independence test of the discretisation and linear solve, not a phase-field calculation: no phase-field or gradient-damage regularisation is claimed to have been reproduced on the external engine.

Mesh refinement (external engine, 16 → 64 grains). Refining the external engine alone from the production 4×4 (16-element) to an 8×8 (64-element) RVE—orientation set regenerated at the same seed 42—lowers the Ti-6Al-4V localisation-onset fracture strains by 7–10% (0.0493 → 0.0444, 0.0345 → 0.0320, 0.0296 → 0.0270 at 0°/45°/90°) while preserving both the $0° > 45° > 90°$ ordering and the localisation-onset criterion, so the external cross-check is mesh-robust in its qualitative content. The AlSi10Mg 0° case moves further and changes its *active criterion*, from the aggregate D_c crossing (0.0564 at 16 elements) to localisation onset (0.0374 at 64 elements)—directly demonstrating the mesh- and criterion-sensitivity that produced the 4×4 disagreement between the two engines: on a coarse RVE neither the fracture-strain magnitude nor the active criterion is converged, which is why only the *sign* and the *orientation trend* are quoted as robust. The in-house counterpart at 64 grains was not carried to convergence in this environment—its damped Newton collapses the strain increment by more than an order of magnitude near localisation at the finer mesh—so this mesh-stability statement is reported for the external engine alone.

Provenance of the external run. The engine was installed with `pip` into a Python 3.12 virtual environment under Windows Subsystem for Linux 2 (Ubuntu-24.04); `scikit-fem` 12.0.2, `numpy` 2.5.3 and `scipy` 1.18.1 resolved with clean TLS. The conda/micromamba route was blocked by an interception proxy presenting a self-signed certificate, `fenics-dolfinx` is absent from PyPI, and the `win-64` conda-forge constraint of Section S9 applies, so `scikit-fem` was the only installable external finite-element engine. The scripts `fullfield_skfem.py` (solver) and `compare_skfem.py` (comparison driver) and their result JSONs `fullfield_skfem_results.json` and `skfem_vs_inhouse_comparison.json` are archived with the code; the mesh-refinement run above reuses the same driver at `FF_NGRID=8` and emits `fullfield_skfem_results_ng8.json`; the wall-clock per case was 19–68 s (4×4) and 65–200 s (8×8).

Fully independent re-implementation on a second finite-element engine (FEniCSx). The `scikit-fem` cross-check above

Table S25: Fracture strain ε_f from the external `scikit-fem` 12 full-field engine against the in-house two-dimensional CPFEM reference under the identical constitutive law, parameters, orientation set and discretisation. $\Delta = (\varepsilon_f^{\mathrm{skfem}} - \varepsilon_f^{\mathrm{inhouse}})/\varepsilon_f^{\mathrm{inhouse}}$.

Material	**Criterion**	$\varepsilon_f^{\mathrm{inhouse}}$	$\varepsilon_f^{\mathrm{skfem}}$	Δ / %
Ti-6Al-4V 0°	localisation onset	0.0555	0.0493	−11.2
Ti-6Al-4V 45°	localisation onset	0.0413	0.0345	−16.3
Ti-6Al-4V 90°	localisation onset	0.0358	0.0296	−17.3
AlSi10Mg 0°	aggregate D_c	0.122	0.0564	−53.8

re-drives the *same* batch explicit constitutive update copied from the in-house solver, so it verifies the assembly and linear solve but not the re-derivation of the physics, and it reuses a Q4 element. To test both the physics and the discretisation, the coupled-damage plane-strain boundary-value problem was re-implemented from scratch in FEniCSx (`dolfinx` 0.11.0): a different library, a different element (continuous P1 triangles with discontinuous DG0 stress and algorithmic tangent integration variables, rather than Q4), and a different solution strategy (a full Newton iteration in which a consistent algorithmic tangent $\partial\sigma_{ij}/\partial\varepsilon_{kl}$ is refreshed at every iteration by forward-differencing the crystal update, assembled into the PETSc block system and solved by a direct LU factorisation, with adaptive strain stepping and a residual-based line search). The crystal update was independently recoded from the single-crystal parameters extracted from the mean-field solver rather than reused verbatim, on the same plane-strain 4×4 RVE, seed 42 16-grain orientation set, anisotropic $\mathbb{C}_4$ stiffness, Voce plus latent hardening, AM-corrected Lemaitre driving force, plastic-strain activation gate and $D_c = 0.42$ fracture criterion.

All three Ti-6Al-4V build orientations again return the localisation-onset criterion and preserve the $0° > 45° > 90°$ ordering. Recomputing the Taylor baseline on the identical strain window, the FEniCSx re-implementation reproduces the main-text negative Taylor-homogenisation bias at every orientation (−29.1%, −38.8% and −45.4% against the in-house −26.5%, −32.6% and −35.8%): the same sign and the same monotone growth with build angle, obtained on an independently coded constitutive update and a different element. The FEniCSx fracture strains are systematically a little lower and the cross-engine gap widens with ψ (−12% at 0° to −33% at 90°). This is the expected consequence of the localisation-onset test on a coarse 16-element RVE: the steeper softening at higher AM constraint ($f_{\mathrm{AM}} = 0.38 \to 0.77 \to 1.66$) drives the loss of equilibrium earlier and makes the onset strain the least mesh- and solver-robust quantity—exactly the caveat carried in the main text and above, where only the *sign* and the *orientation trend* are claimed as converged. The result therefore strengthens the independence evidence two ways over `scikit-fem`: a genuinely different discretisation *and* an independently re-derived constitutive update both leave the negative-bias conclusion and its orientation trend intact.

Provenance of the FEniCSx run. Where the Windows production environment cannot host `fenics-dolfinx` (no `win-64` build at Python 3.13, Section S9), the solver was built in a separate Linux `conda-forge` (`linux-64`) environment under Windows Subsystem for Linux 2 (Ubuntu-24.04)—precisely the second conda environment pinned below the production interpreter version that Section S9 anticipated as the only way to run FEniCSx here. The solver script `verify_fenicsx_ti64_2d.py` and its result file `verify_fenicsx_ti64_2d.json` are archived with the code; the elastic-limit self-check returns $\sigma_{11} = 131\,\mathrm{MPa}$ at $\varepsilon_{11} = 10^{-3}$, between the uniaxial and the fully constrained plane-strain bounds, and the peak flow stress is $\approx 1272\,\mathrm{MPa}$ at $\psi = 0$, consistent with the Ti-6Al-4V reference strength.

Table S26: Fracture strain ε_f from the independent FEniCSx 0.11 full-field re-implementation against the in-house two-dimensional CPFEM reference under the identical parameters, orientation set and RVE. $\Delta = (\varepsilon_f^{\text{fenicsx}} - \varepsilon_f^{\text{inhouse}})/\varepsilon_f^{\text{inhouse}}$. All three cases activate the localisation-onset criterion.

Material	**Criterion**	$\varepsilon_f^{\text{inhouse}}$	$\varepsilon_f^{\text{fenicsx}}$	Δ / %
Ti-6Al-4V 0°	localisation onset	0.0555	0.0488	−12.0
Ti-6Al-4V 45°	localisation onset	0.0413	0.0316	−23.4
Ti-6Al-4V 90°	localisation onset	0.0358	0.0241	−32.8

S14. Strict frozen-parameter blind prediction on external literature (zero refit)

This section consolidates, under a single *zero-refit* rule, the blind predictions that the main text reports piecemeal. Every model parameter is fixed once by the nine-case primary calibration (the 0°/45°/90° Ti-6Al-4V, 316L and AlSi10Mg set of the main text, source id `v12_original`): the single adjustable damage strength per material takes its published value $S_0 = 0.32\,\text{MPa}$ (Ti-6Al-4V), $3.2\,\text{MPa}$ (316L) and $0.20\,\text{MPa}$ (AlSi10Mg), and all other constants are the a-priori literature/microstructure values of the main text. These parameters are then applied *directly* to tensile data drawn from other publications; no external point is used to re-fit anything — there is no grid search, no refinement and no per-fold recalibration. This is stricter than the leave-one-source-out statistic of S11 (where each fold re-calibrates S_0 on its own training sources) and than the main-text 316L check (where S_0 was re-fit on the held source's 0° fold alone); here the same frozen S_0 is reused for every blind point of a material. The result is produced by the archived driver `blind_predict_frozen.py` and `blind_predict_frozen_results.json`. Heat-treated (HIP/solution/aged) and IN718 rows are excluded because they are outside the nine-case domain and cannot be reached without a material or heat-treatment transfer; only as-built points on the three calibrated materials enter.

Headline result (nine in-domain points). Under the frozen parameters the nine-point mean absolute fracture-strain error is 66.5% (median 32.7%, maximum 360.8%), with 1/9 within 10%, 3/9 within 20% and 4/9 within 30%; the mean UTS error is 18.0% (3/9 within 10%). The per-material picture is the informative one: Ti-6Al-4V (Sun) averages 22.8% and AlSi10Mg (Awd) 38.8% — both *shape* mismatches in which the frozen polarity (0° most ductile) agrees with the measurement but the monotonic projection over-degrades the transverse fold against a V-shaped datum — while 316L (Hitzler) averages 137.8% because that source is genuinely *polarity-flipped* (its 90° is the most ductile, opposite to the calibration set), which no scalar S_0 can recover. The single 360.8% entry (0°, Hitzler) is the main text's already-flagged low-ductility outlier (milled specimens, UTS below the nominal band); dropping it leaves the other two Hitzler folds averaging 26.3%, so the reversal — not the one anomalous point — drives the 316L error.

Relation to the main-text numbers. The frozen Ti-6Al-4V (22.8%) and AlSi10Mg (38.8%) averages reproduce the main-text cross-source values exactly, because those two checks already applied the calibration S_0 directly. The 316L value differs by construction: the main text re-fits S_0 on the Hitzler 0° fold ($S_0^* = 1.0\,\text{MPa}$, 80.6%), whereas the strict zero-refit rule here keeps the calibration $S_0 = 3.2\,\text{MPa}$ and returns 137.8%. The two bracket the same conclusion — a transferred S_0 does not cross sources — and the frozen figure is the more conservative statement of it, reported here so that no blind number depends on a fold-specific refit.

Supplementary out-of-band check. The two NIST mds2-3402 as-built Ti-6Al-4V rows have UTS 1294–1303 MPa, above the nominal band the calibration set was compiled over; the frozen model under-predicts their (much

Table S27: Strict frozen-parameter blind prediction on external as-built literature (one complete 0°/45°/90° trio per material). Parameters frozen at the nine-case calibration, no refit. $\varepsilon_f^{\mathrm{exp}}$ and UTS are measured; predictions are at the frozen per-material S_0. Errors are relative to measurement. Source names follow the main text (Sun et al. 2024; Hitzler et al. 2017; Awd et al. 2018). The last two rows are a supplementary out-of-band-strength Ti-6Al-4V check (NIST mds2-3402 as-built, HT0).

				ε_f			UTS / MPa		
Material	**Source**	ψ [°]	S_0	exp.	pred.	err /%	exp.	pred.	err /%
Ti-6Al-4V	Sun 2024	0	0.32	0.0934	0.0723	22.6	1165.0	959.9	17.6
Ti-6Al-4V	Sun 2024	45	0.32	0.0533	0.0541	1.5	1299.0	952.4	26.7
Ti-6Al-4V	Sun 2024	90	0.32	0.0789	0.0439	44.3	1186.0	950.3	19.9
316L	Hitzler 2017	0	3.20	0.1176	0.5419	360.8	512.0	620.6	21.2
316L	Hitzler 2017	45	3.20	0.3256	0.3902	19.9	699.0	501.3	28.3
316L	Hitzler 2017	90	3.20	0.3324	0.2236	32.7	634.4	411.8	35.1
AlSi10Mg	Awd 2018	0	0.20	0.0830	0.0688	17.1	351.8	351.7	0.0
AlSi10Mg	Awd 2018	45	0.20	0.0570	0.0373	34.7	367.8	349.8	4.9
AlSi10Mg	Awd 2018	90	0.20	0.0810	0.0286	64.7	379.6	349.5	7.9
Supplementary — Ti-6Al-4V, out-of-band strength (reported separately, not in the headline average)									
Ti-6Al-4V	NIST mds2 HT0	0	0.32	0.1680	0.0723	57.0	1302.7	959.9	26.3
Ti-6Al-4V	NIST mds2 HT0	90	0.32	0.1928	0.0439	77.2	1294.2	950.3	26.6

higher) ductility by 57% and 77% and their strength by $\approx 26\%$. They are kept out of the headline average and reported separately, because a strength level outside the calibration band is a process-window extrapolation rather than an orientation test; their behaviour is the same magnitude-not-direction failure class the main text identifies for Inconel 718.

S15. Additional figures and tables

The following figures and tables support the main text.

Table S28: Calibration data: experimental fracture strain ε_f for three AM alloys at three build orientations (literature values; see the main text for sources and test conditions).

Material	ε_f (0°)	ε_f (45°)	ε_f (90°)
Ti-6Al-4V	0.080	0.063	0.048
316L SS	0.370	0.310	0.255
AlSi10Mg	0.056	0.047	0.037

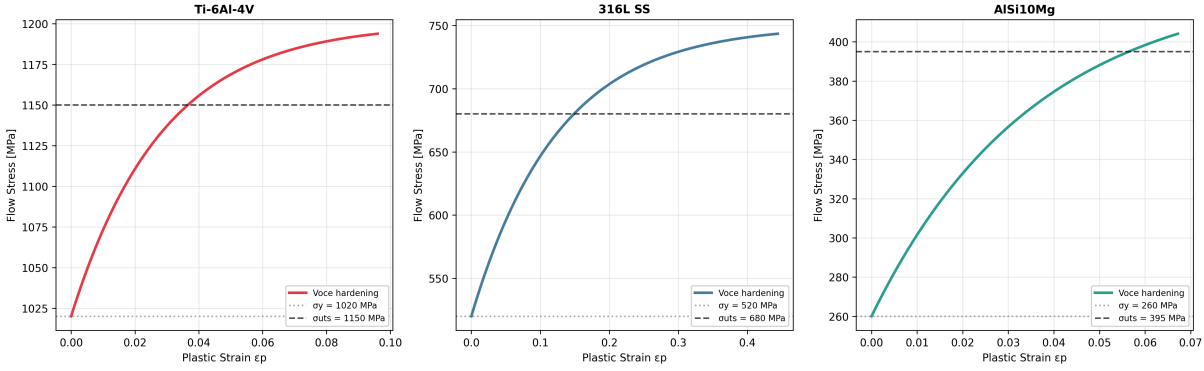


Figure S5: Voce hardening curves of the three AM alloys with the literature-fixed parameters of main-text parameter table. The curves are not fitted to fracture data; the flow response follows from the fixed hardening constants.

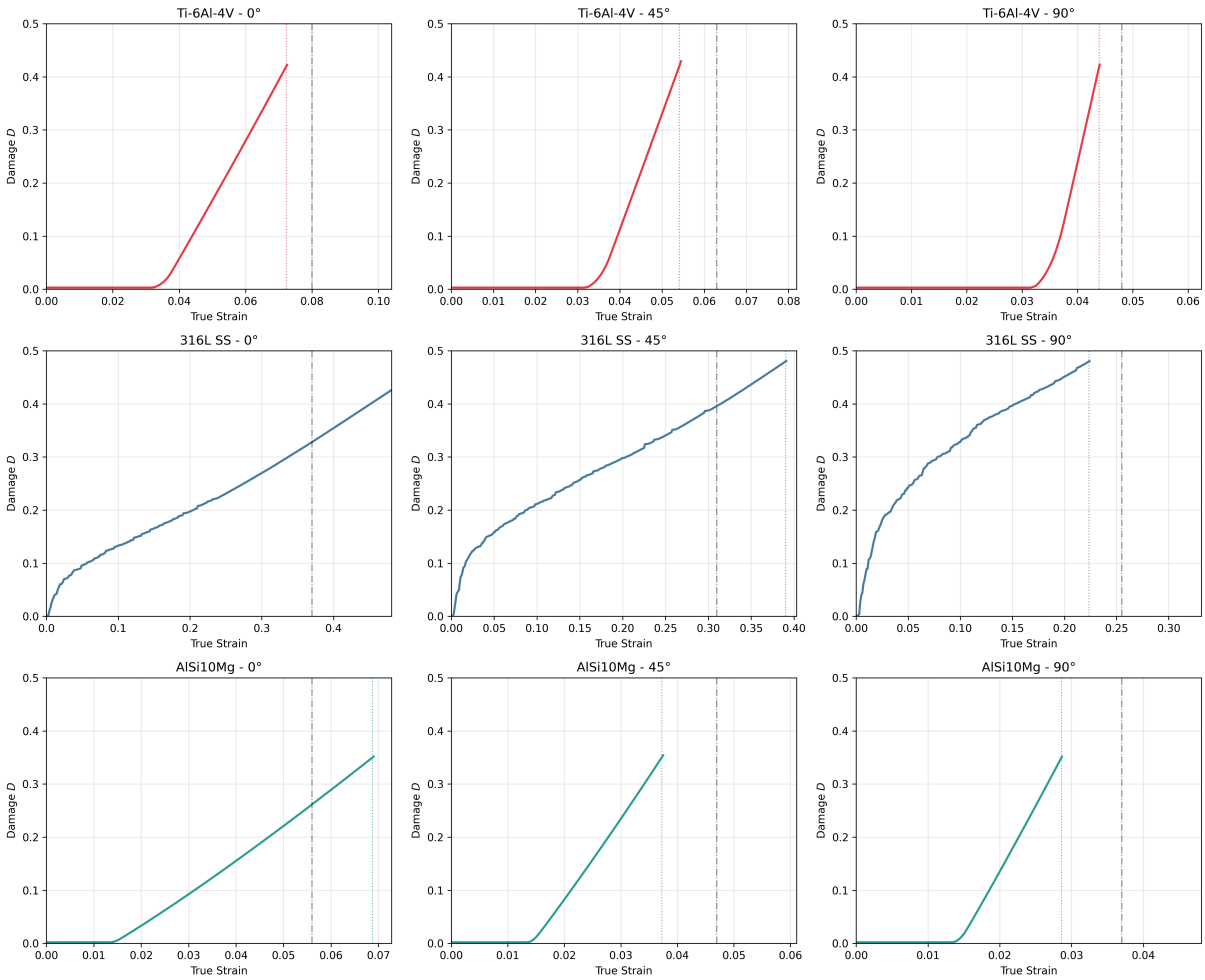


Figure S6: Damage evolution trajectories predicted by the modified CDM model for all nine calibration cases. The horizontal dashed red line marks the critical damage D_c used to declare fracture; the orientation-dependent damage acceleration is captured from 0° to 90°.

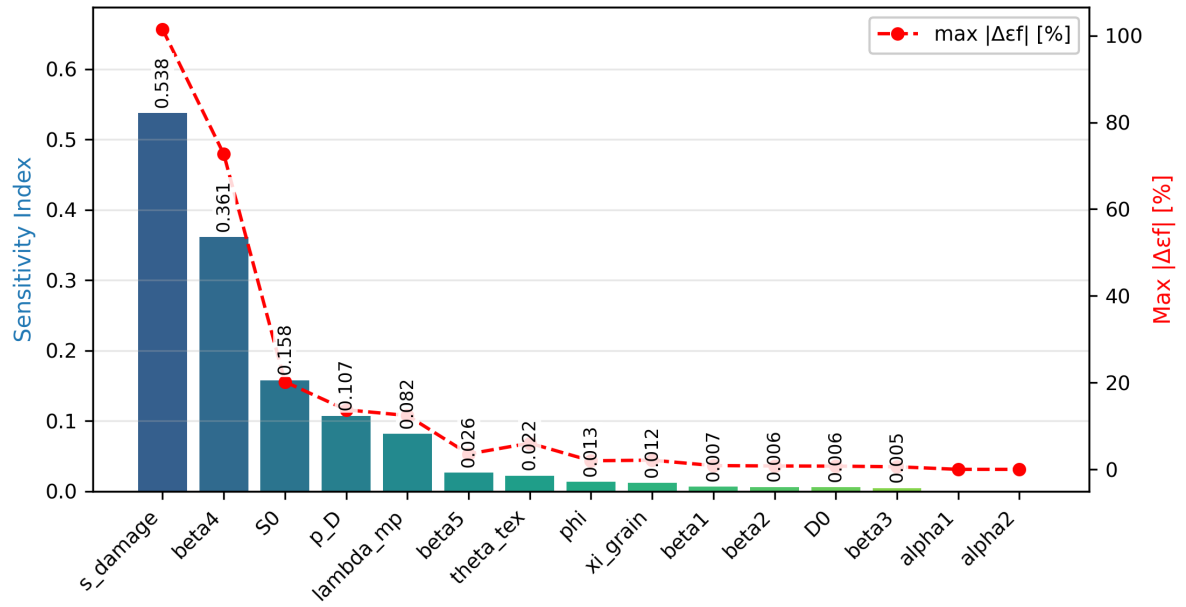


Figure S7: One-at-a-time (OAT) sensitivity of the predicted fracture strain to ±40% variations of each damage-model parameter (Ti-6Al-4V, 0°, production leakage-free parameters).

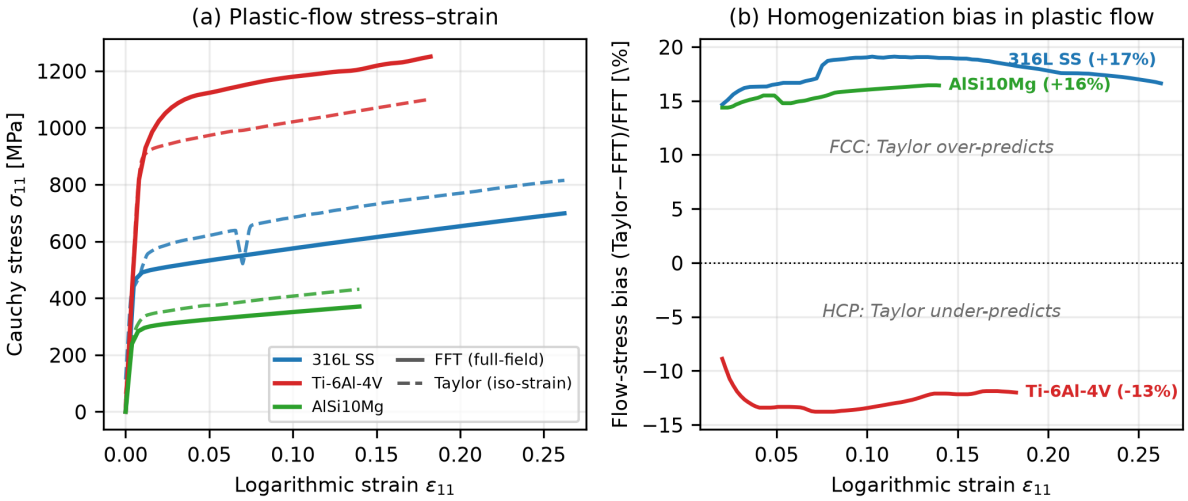


Figure S8: Taylor iso-strain (dashed) versus DAMASK spectral-FFT full-field (solid) *plastic-flow* response, damage-free, loading along the build direction. (a) True (Cauchy) stress against logarithmic strain for the three alloys; the surrogate misrepresents the flow-stress level throughout the plastic branch, not only at the UTS. (b) Resulting flow-stress bias $(\sigma^{\mathrm{Taylor}} - \sigma^{\mathrm{FFT}})/\sigma^{\mathrm{FFT}}$ as a function of plastic strain: the FCC alloys are over-predicted (+15 to +19%) and the HCP Ti-6Al-4V under-predicted (−11 to −14%), and the monotone drift of each curve with strain is the hardening-rate bias (Table S30).

Table S29: Nine-case calibration results of the modified CDM model under the production leakage-free parameterization (single S_0 per material, literature g_s, directional slip-resistance correction $k_g = 0.15$, no orientation multipliers): experimental (Exp.) and predicted (Pred.) fracture strain ε_f and UTS, with percentage errors. These are *calibration* errors on data used to fit S_0; independent blind prediction is reported in main-text leave-one-orientation-out table.

Material	**Ori.**	**Exp.**	**Pred.**	**Error [%]**	**Pass**
Ti-6Al-4V	0°	0.080	0.0723	9.64	✓
	45°	0.063	0.0541	14.16	–
	90°	0.048	0.0439	8.47	✓
316L SS	0°	0.370	0.5419	46.45	–
	45°	0.310	0.3902	25.89	–
	90°	0.255	0.2236	12.32	–
AlSi10Mg	0°	0.056	0.0688	22.80	–
	45°	0.047	0.0372	20.75	–
	90°	0.037	0.0286	22.69	–
Average error				**20.35%**	**2/9**

Table S30: Plastic-flow bias of the Taylor iso-strain surrogate against the DAMASK spectral-FFT full-field reference (damage-free, build-direction loading, common plastic window $\varepsilon \in [0.02, \varepsilon_{\max}]$). Flow-stress bias is the mean relative stress deviation; hardening-rate bias is the deviation of the secant $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ over the same window.

Material	**UTS bias [%]**	**Flow-stress bias [%]**	**Hardening-rate bias [%]**
316L (FCC)	+16.7	+17.4	+21.7
AlSi10Mg (FCC)	+16.5	+15.6	+26.2
Ti-6Al-4V (HCP)	−12.0	−12.7	−26.2

Table S31: Coupled-damage Taylor-homogenization bias on the fracture strain, from the in-house full-field CPFEM references under the identical production leakage-free parameters, orientation set and damage law. Bias $= (\varepsilon_f^{\mathrm{ff}} - \varepsilon_f^{\mathrm{Taylor}})/\varepsilon_f^{\mathrm{Taylor}}$: negative ⇒ the iso-strain surrogate over-estimates the fracture strain, positive ⇒ it under-estimates it. The sign is set by the kinematic constraint and the magnitude by the failure criterion; detail and figures in Supplementary Section S1.

Reference	**Constraint**	**Criterion**	**Material / orientation: bias**
2D CPFEM	plane strain	localisation onset	Ti-6Al-4V 0°/45°/90°: −26.5/ − 32.6/ − 35.8%
2D CPFEM	plane strain	localisation onset	AlSi10Mg 0°: +5.3%
3D CPFEM	free 3D	aggregate D_c	Ti-6Al-4V 0°/45°/90°: +92.7/ + 61.6/ + 49.8%
3D CPFEM	free 3D	localisation onset	Ti-6Al-4V 0°: +64.9%
3D CPFEM (nonlocal)	free 3D	aggregate D_c	Ti-6Al-4V 0°: +105 to +110% (mesh-conv. <1 pp)